Nonparametric Modelling

Bingchen Xie

Disclaimer:

These slides are my own supplementary resource and are not official course materials. Any mistakes are my own.

Please refer to the solutions posted on **Moodle** as the official source.

Question 1

Question 1. Expressiveness of Trees

Give decision trees to represent the following Boolean functions, where the variables `A`, `B`, `C` and `D` have values `t` or `f`, and the class value is either `True` or `False`. Can you observe any effect of the increasing complexity of the functions on the form of their expression as decision trees ?

(a) $A \wedge \neg B$

(b) $A \vee [B \wedge C]$

(c) A XOR B

(d) $[A \wedge B] \vee [C \wedge D]$

Recap: Logic

Question 1. Expressiveness of Trees

Give decision trees to represent the following Boolean functions, where the variables `A`, `B`, `C` and `D` have values `t` or `f`, and the class value is either `True` or `False`. Can you observe any effect of the increasing complexity of the functions on the form of their expression as decision trees ?

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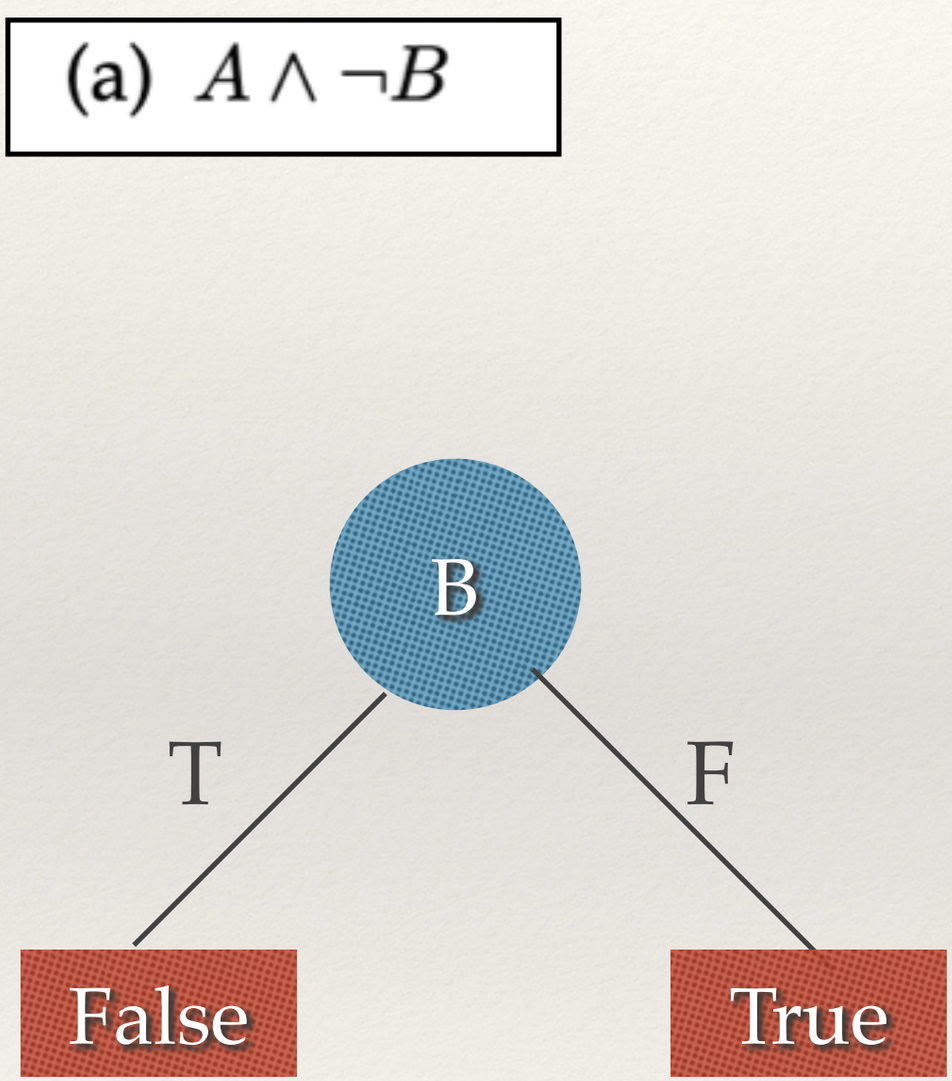
(c) A XOR B

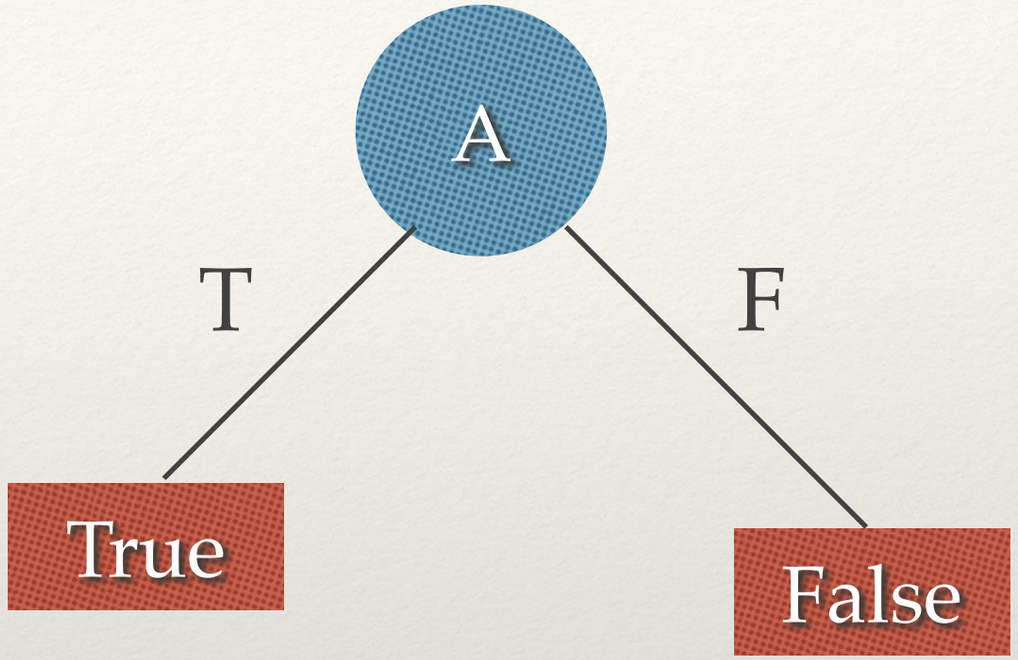
(d) $[A \wedge B] \vee [C \wedge D]$

1. **Commutative laws:**
   $p \wedge q \equiv q \wedge p$
   $p \vee q \equiv q \vee p$
2. **Associative laws:**
   $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
   $(p \vee q) \vee r \equiv p \vee (q \vee r)$
3. **Distributive laws:**
   $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
   $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
4. **Identity laws:**
   $p \wedge t \equiv p$
   $p \vee f \equiv p$
5. **Negation laws:**
   $p \vee \neg p \equiv t$
   $p \wedge \neg p \equiv f$
6. **Double negation law:**
   $\neg(\neg p) \equiv p$
7. **Idempotent laws:**
   $p \wedge p \equiv p$
   $p \vee p \equiv p$
8. **Universal bound laws:**
   $p \vee t \equiv t$
   $p \wedge f \equiv f$
9. **De Morgan's laws:**
   $\neg(p \wedge q) \equiv \neg p \vee \neg q$
   $\neg(p \vee q) \equiv \neg p \wedge \neg q$
10. **Absorption laws:**
    $p \vee (p \wedge q) \equiv p$
    $p \wedge (p \vee q) \equiv p$
11. **Negations of t and f:**
    $\neg t \equiv f$
    $\neg f \equiv t$

Q1a Analyze

(a) $A \wedge \neg B$

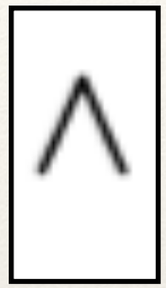


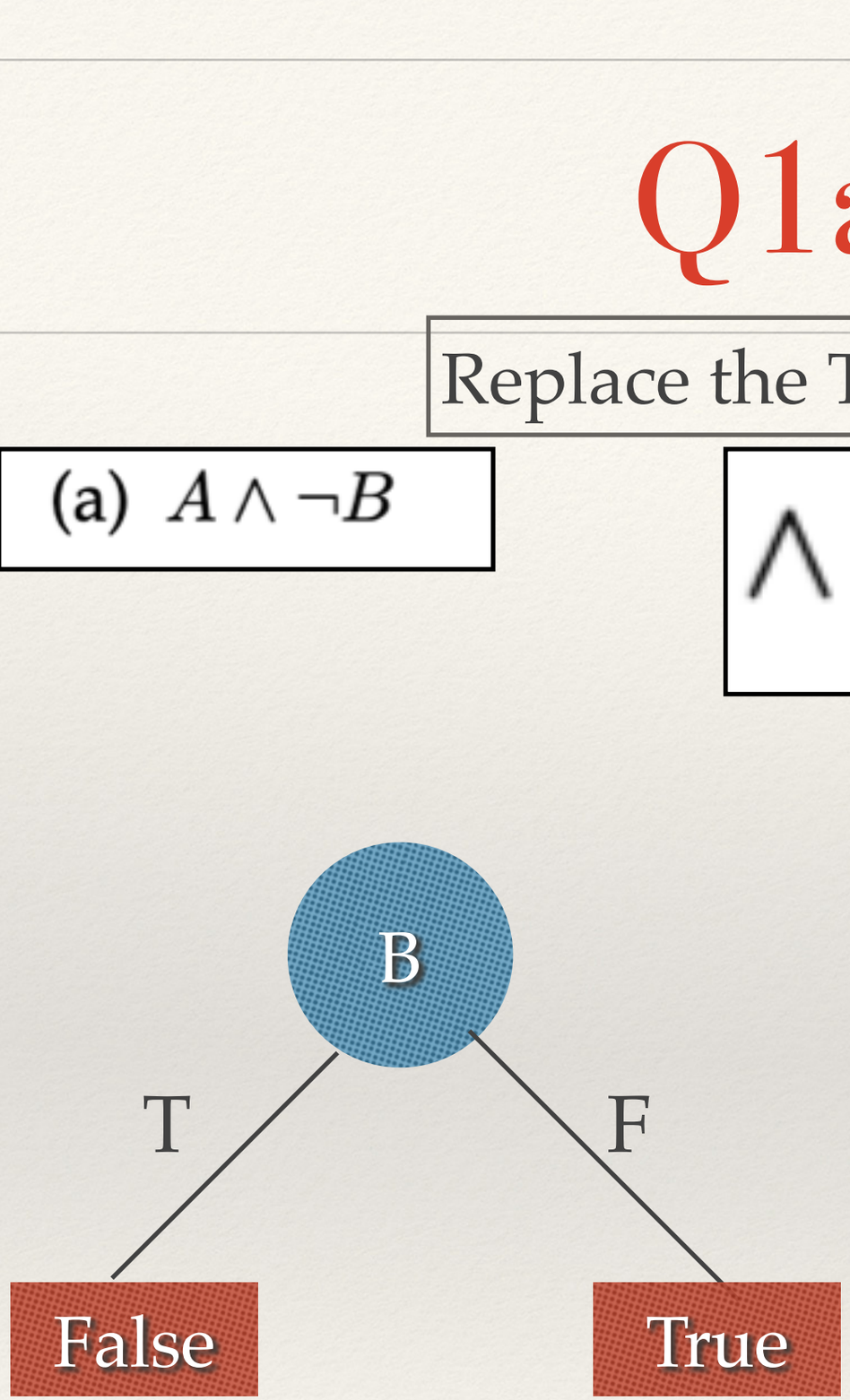


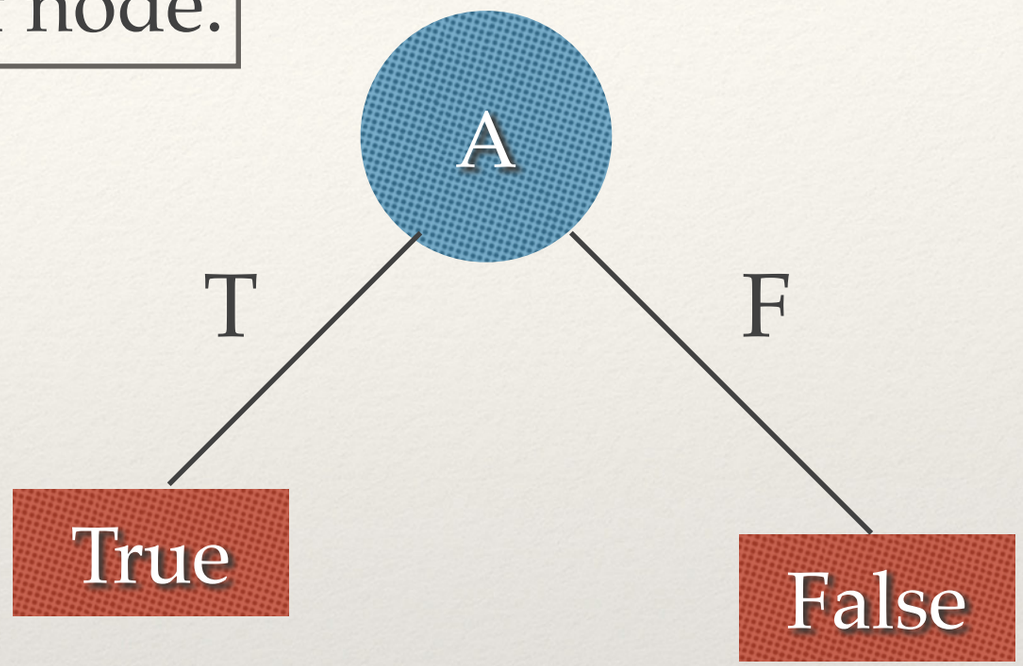
Q1a Analyze

Replace the True leaf node.

(a) $A \wedge \neg B$



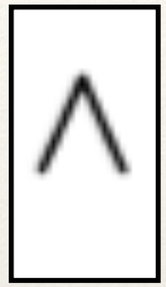


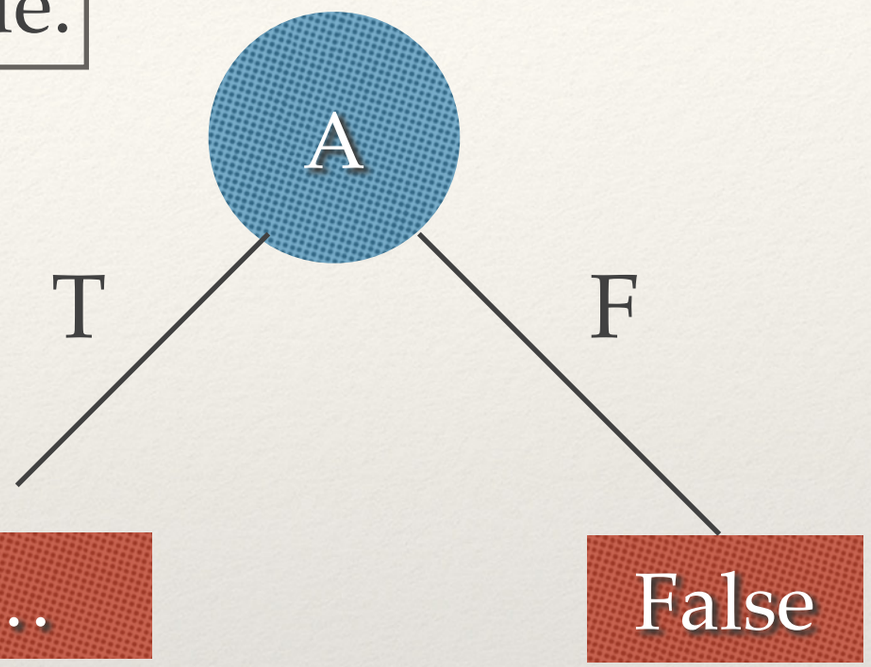


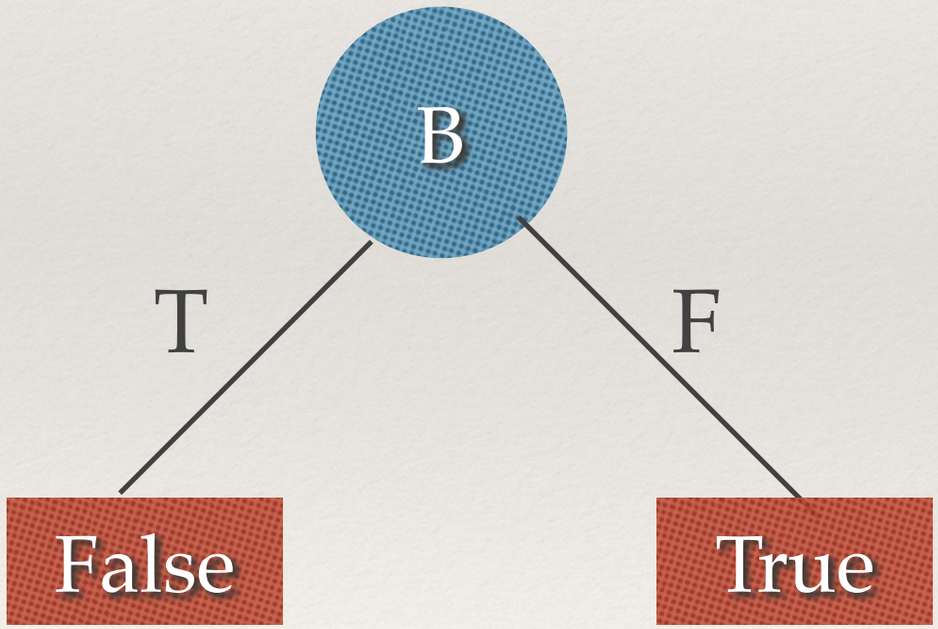
Q1a Analyze

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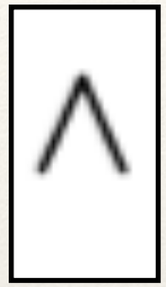


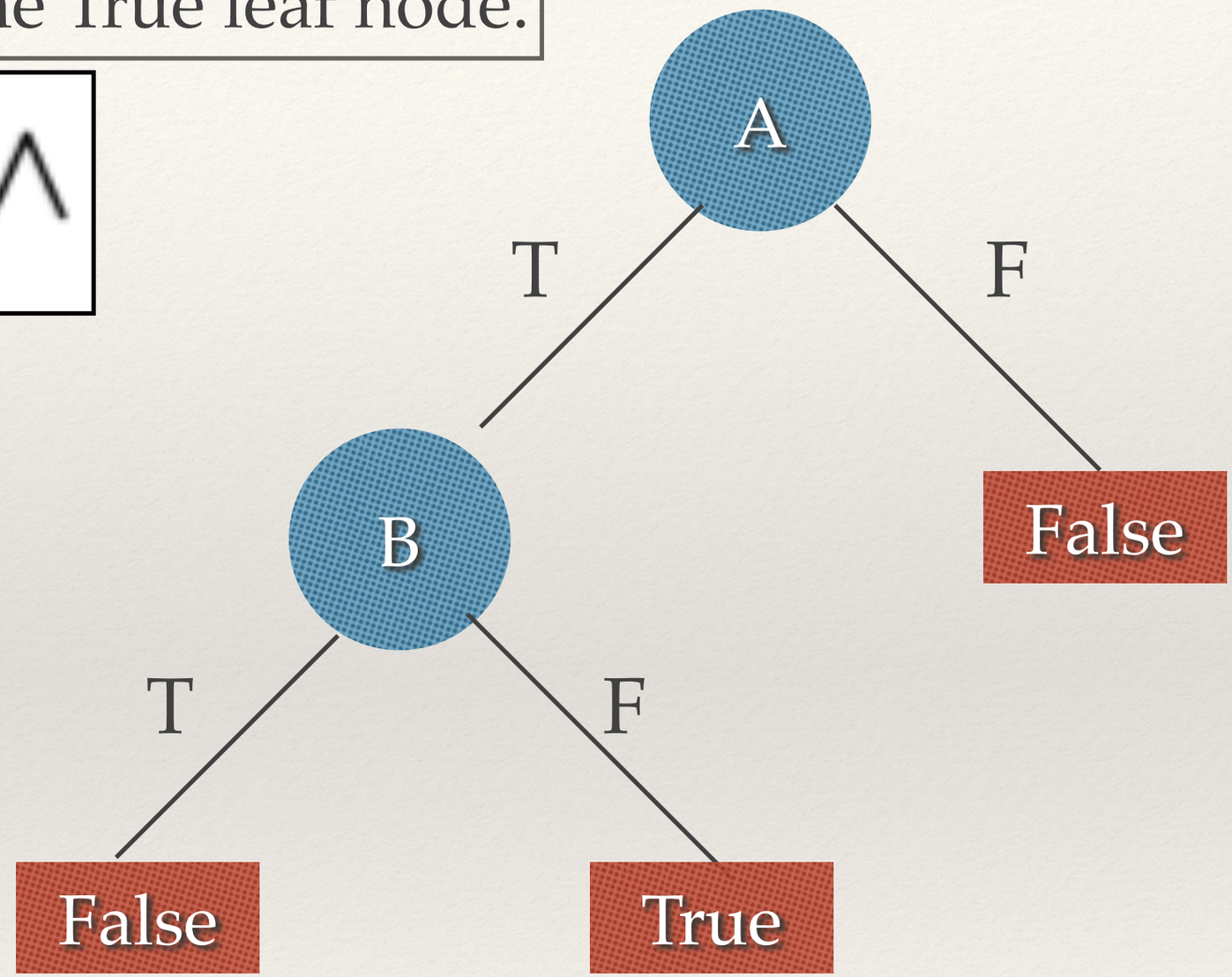


Q1a Analyze

Replace the True leaf node.

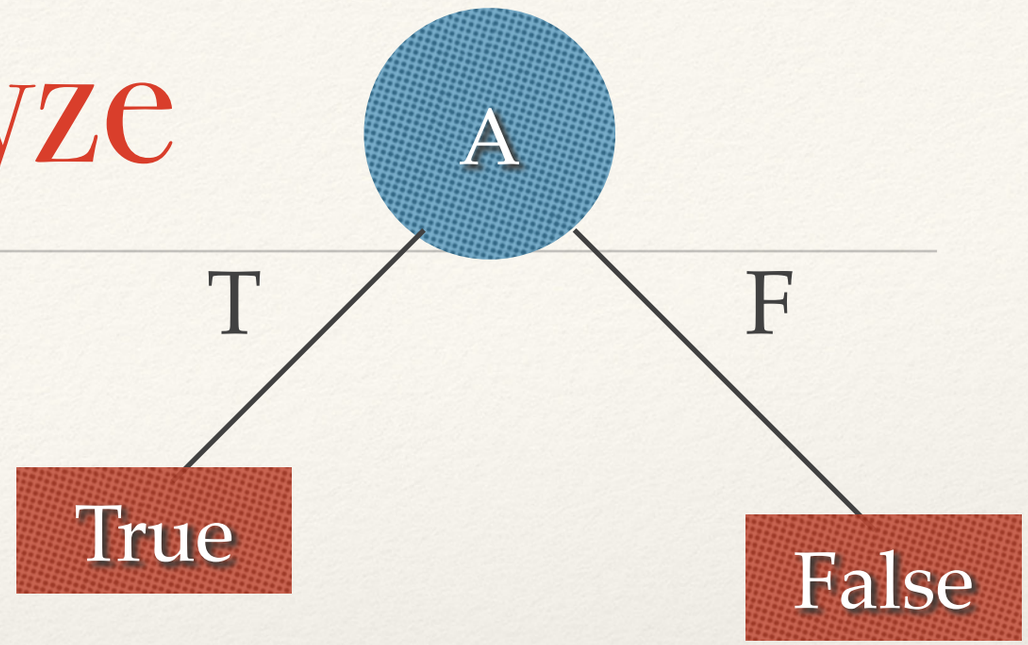
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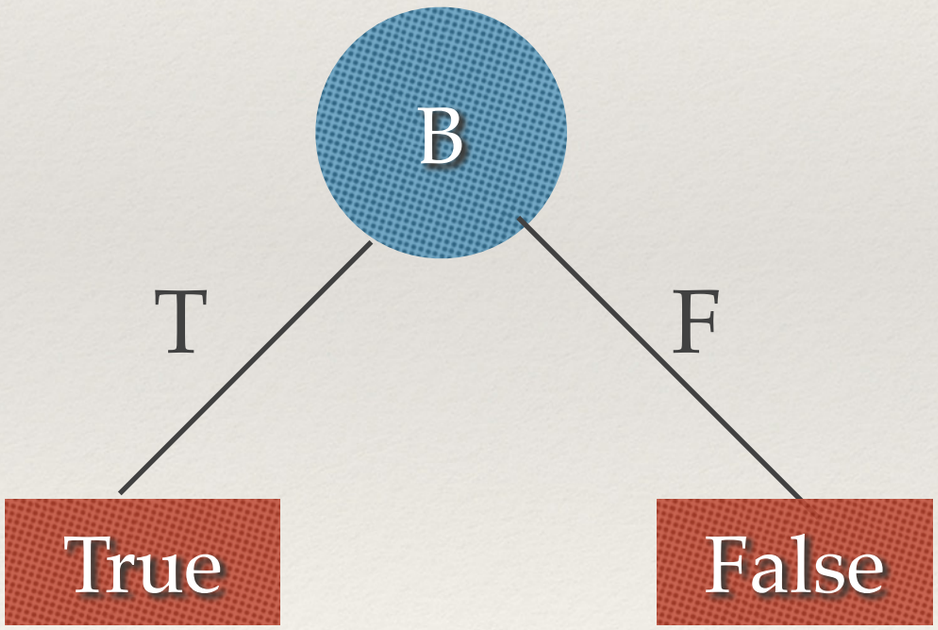


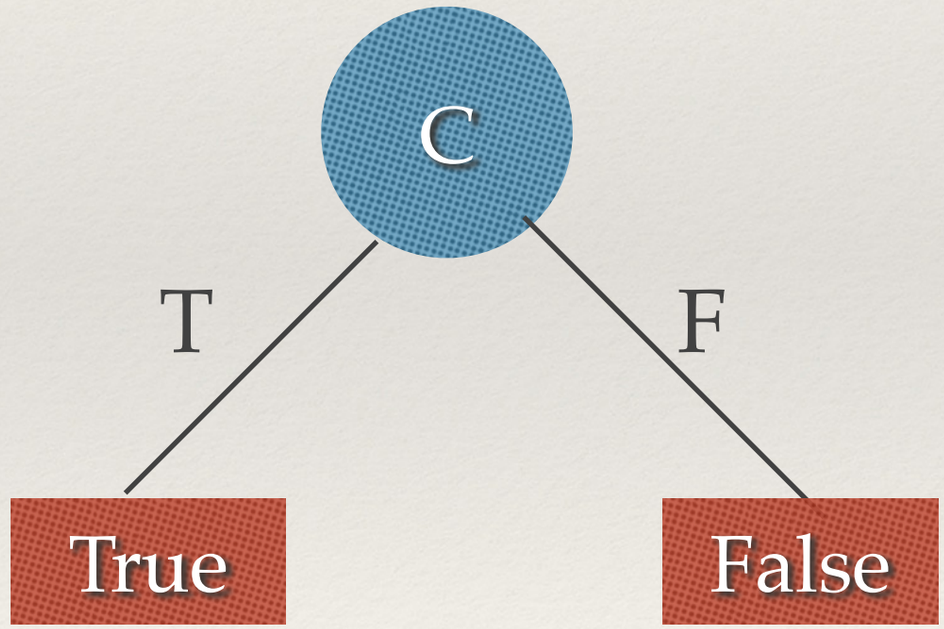


Q1b Analyze

(b) $A \vee [B \wedge C]$





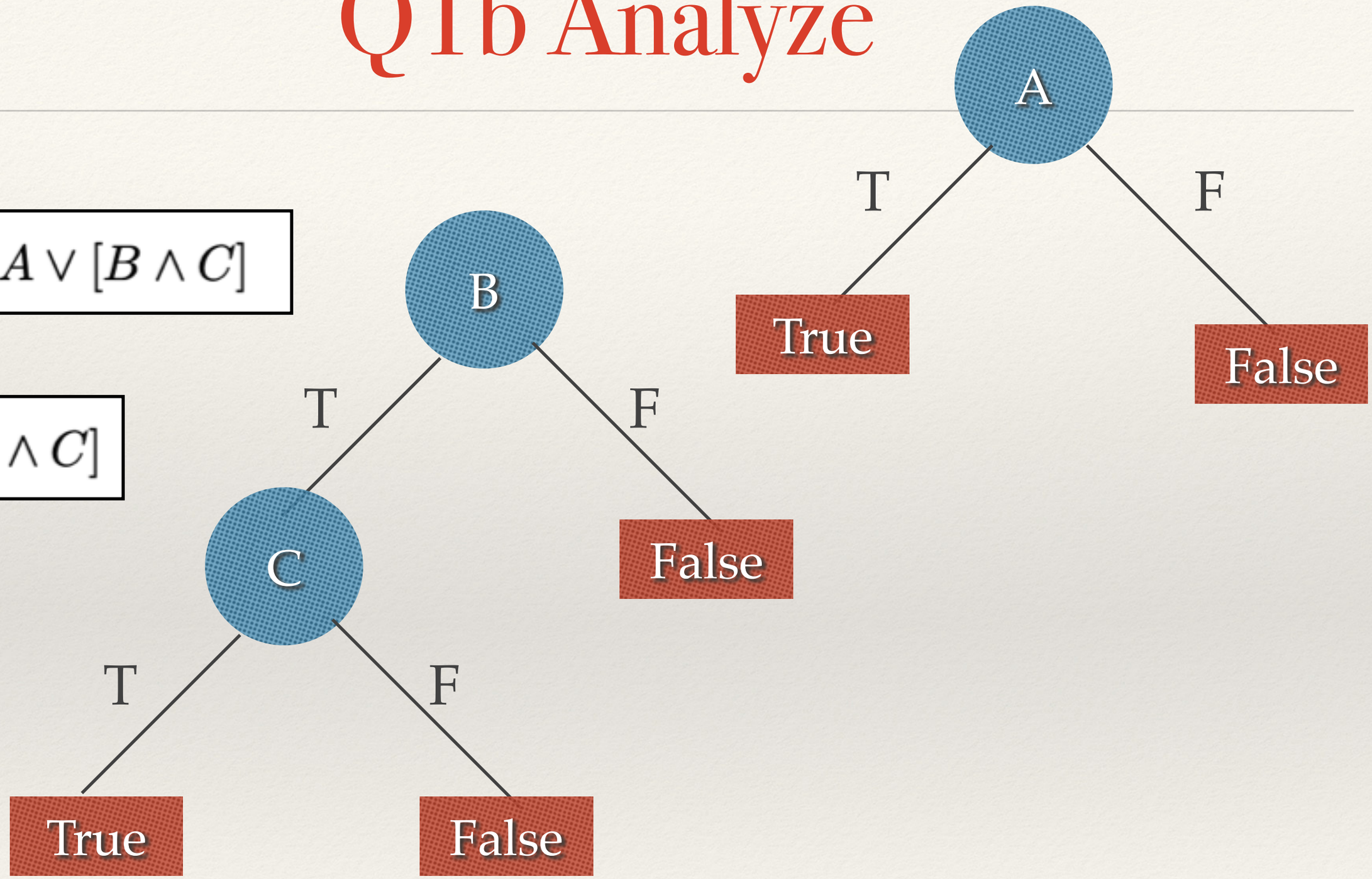


$[B \wedge C]$

Q1b Analyze

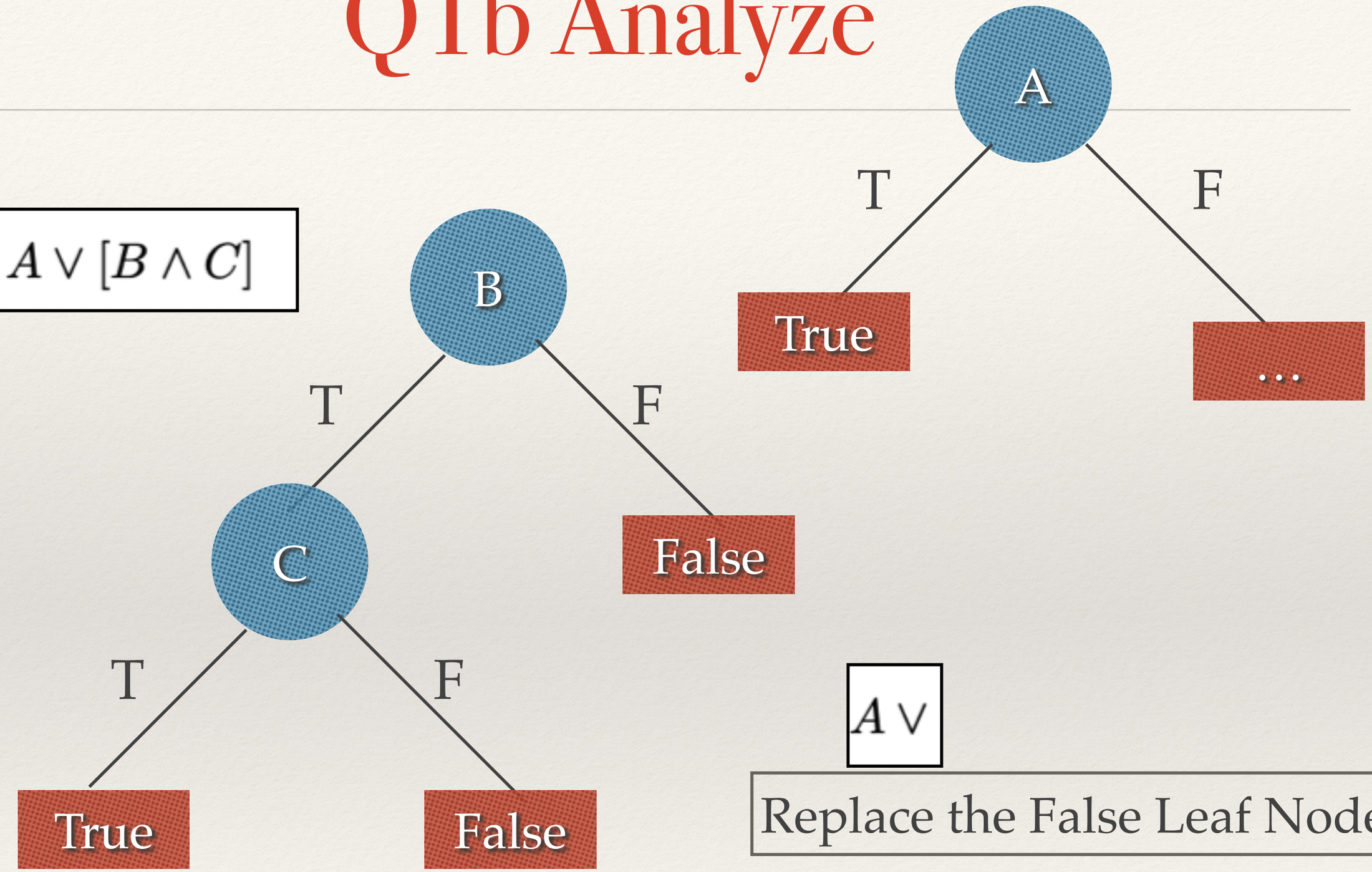
(b) $A \vee [B \wedge C]$

$[B \wedge C]$



Q1b Analyze

(b) $A \vee [B \wedge C]$



$A \vee$

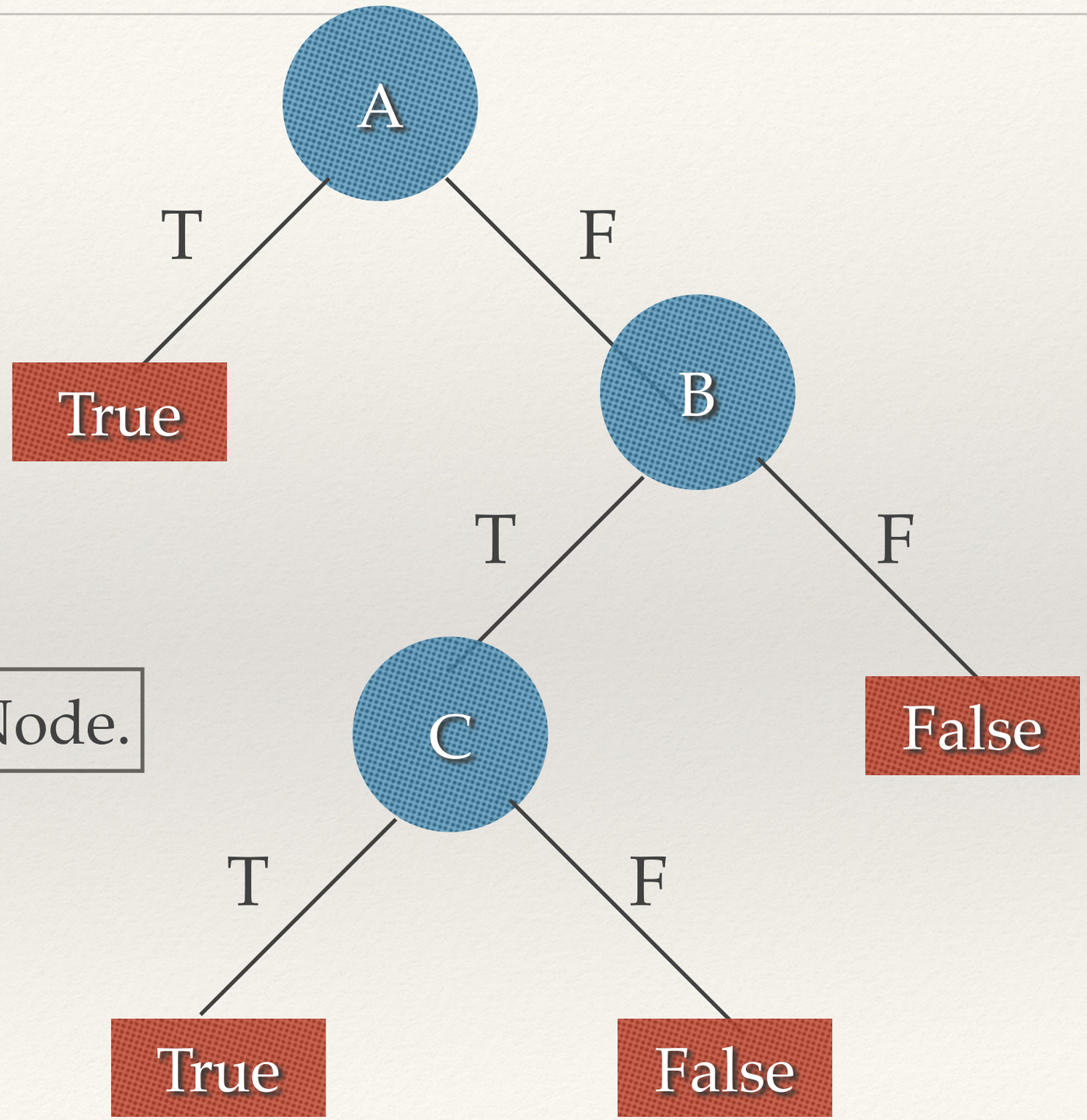
Replace the False Leaf Node.

Q1b Analyze

(b) $A \vee [B \wedge C]$

$A \vee$

Replace the False Leaf Node.



Q1c Analyze

(c) A XOR B

$$A \oplus B \equiv (A \vee B) \wedge \neg(A \wedge B)$$

Both True==> False

Both False ==> False

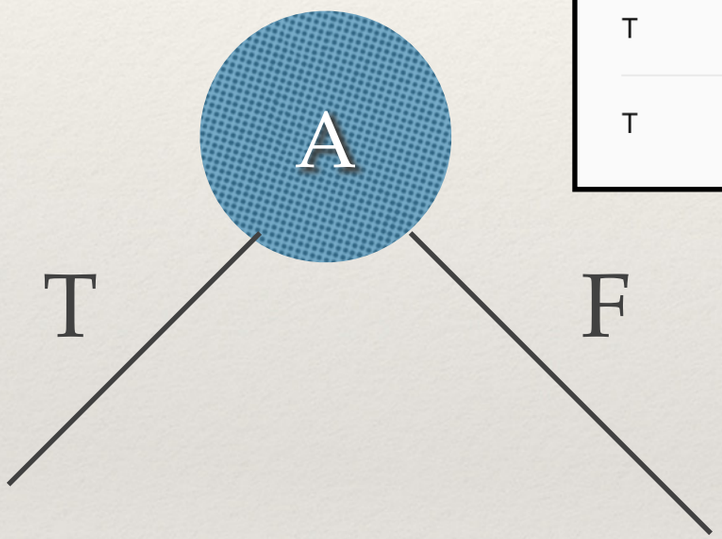
One True, One False ==> True

A	B	$A \oplus B$
F	F	F
F	T	T
T	F	T
T	T	F

Q1c Analyze

(c) A XOR B

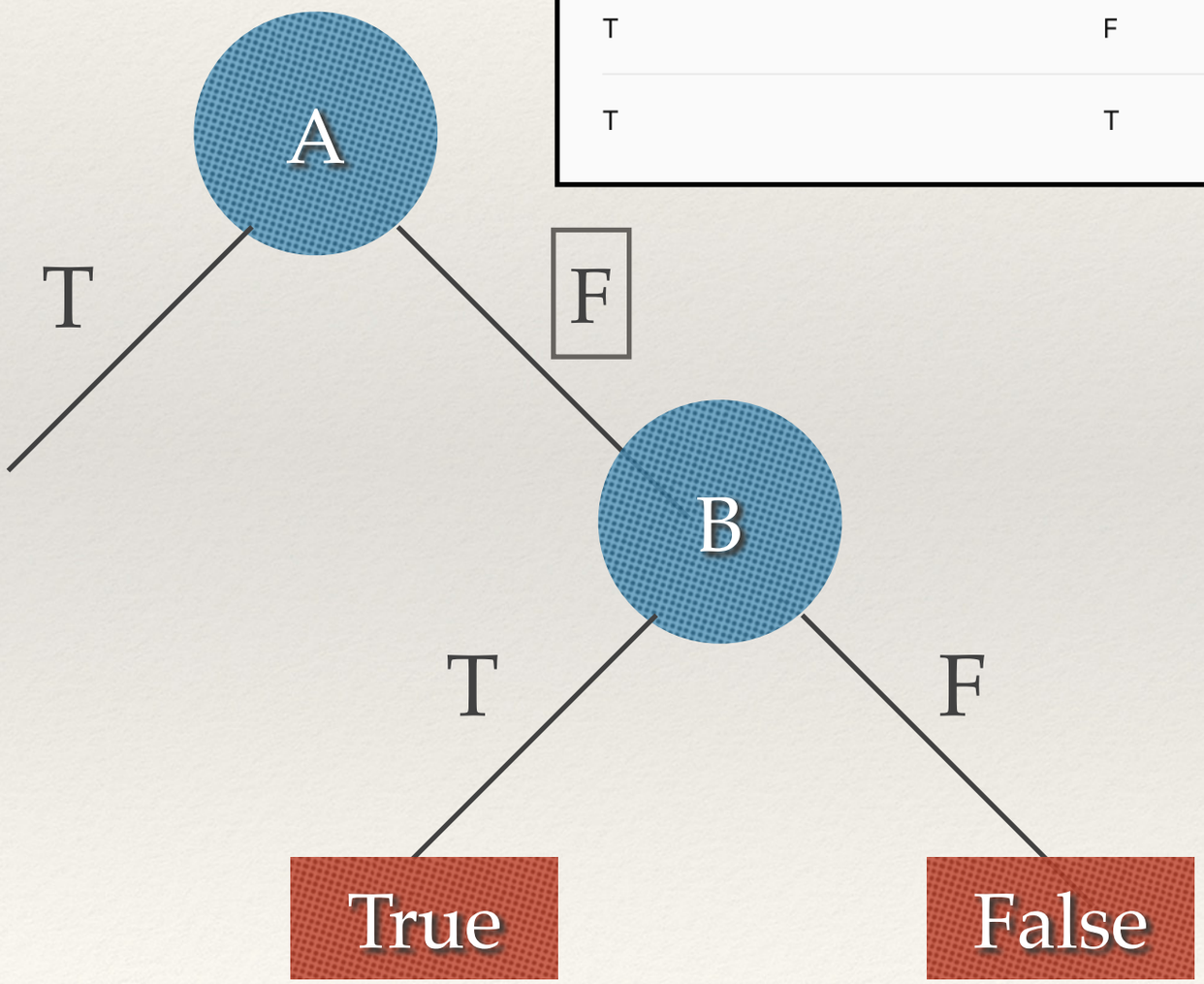
A	B	$A \oplus B$
F	F	F
F	T	T
T	F	T
T	T	F



Q1c Analyze

(c) A XOR B

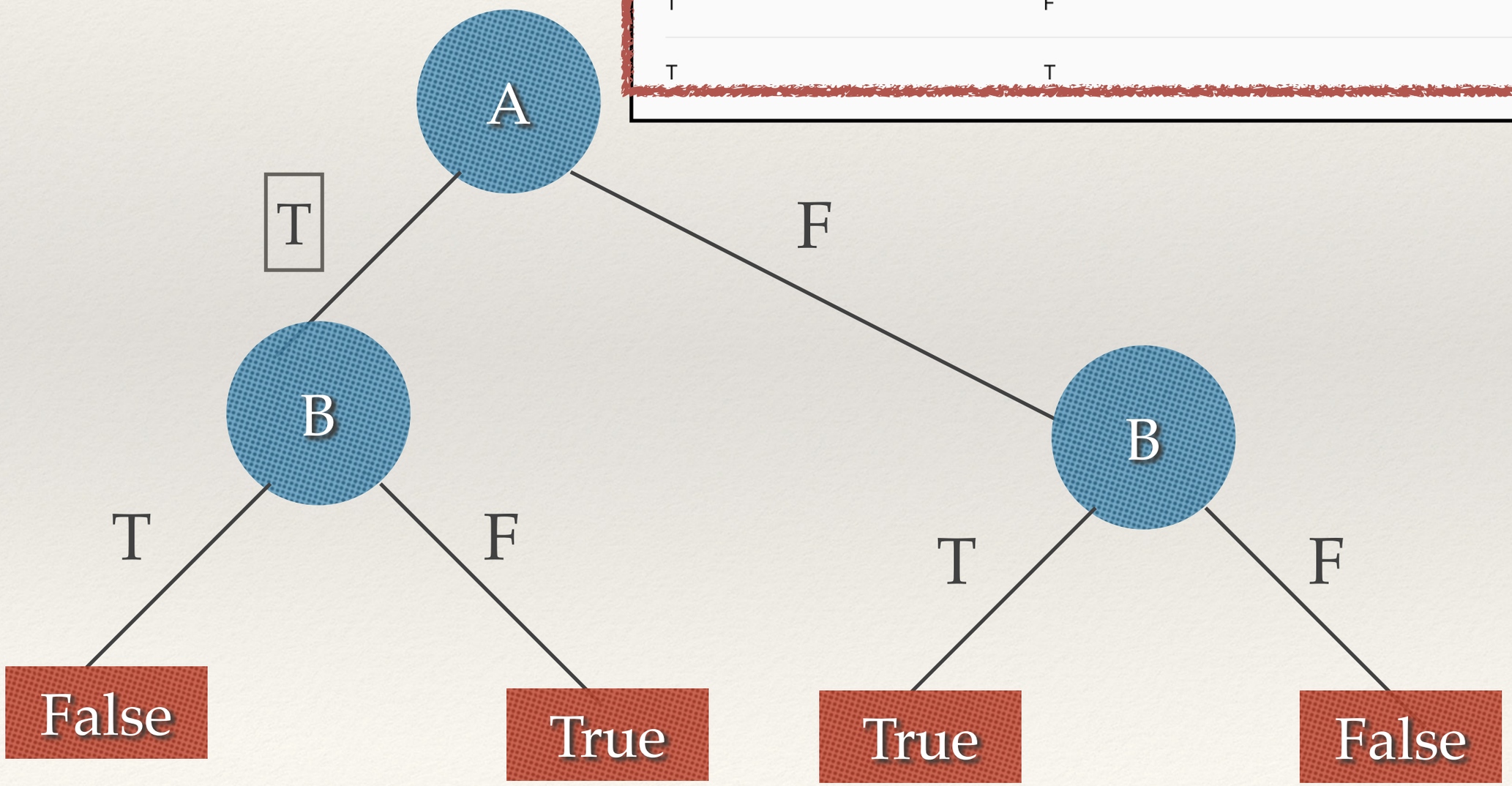
A	B	$A \oplus B$
F	F	F
F	T	T
T	F	T
T	T	F



Q1c Analyze

(c) A XOR B

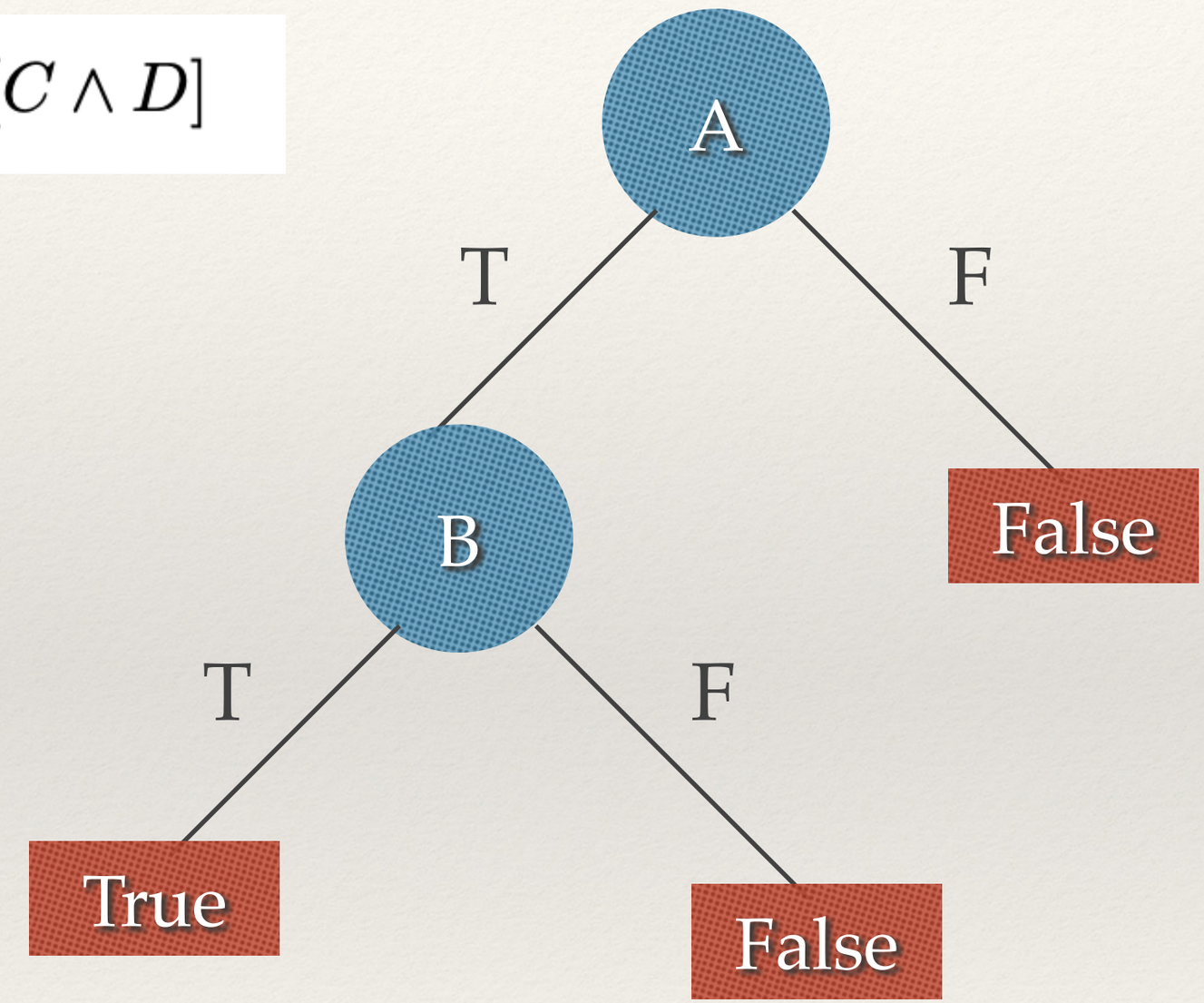
A	B	$A \oplus B$
F	F	F
F	T	T
T	F	T
T	T	F



Q1d Analyze

(d) $[A \wedge B] \vee [C \wedge D]$

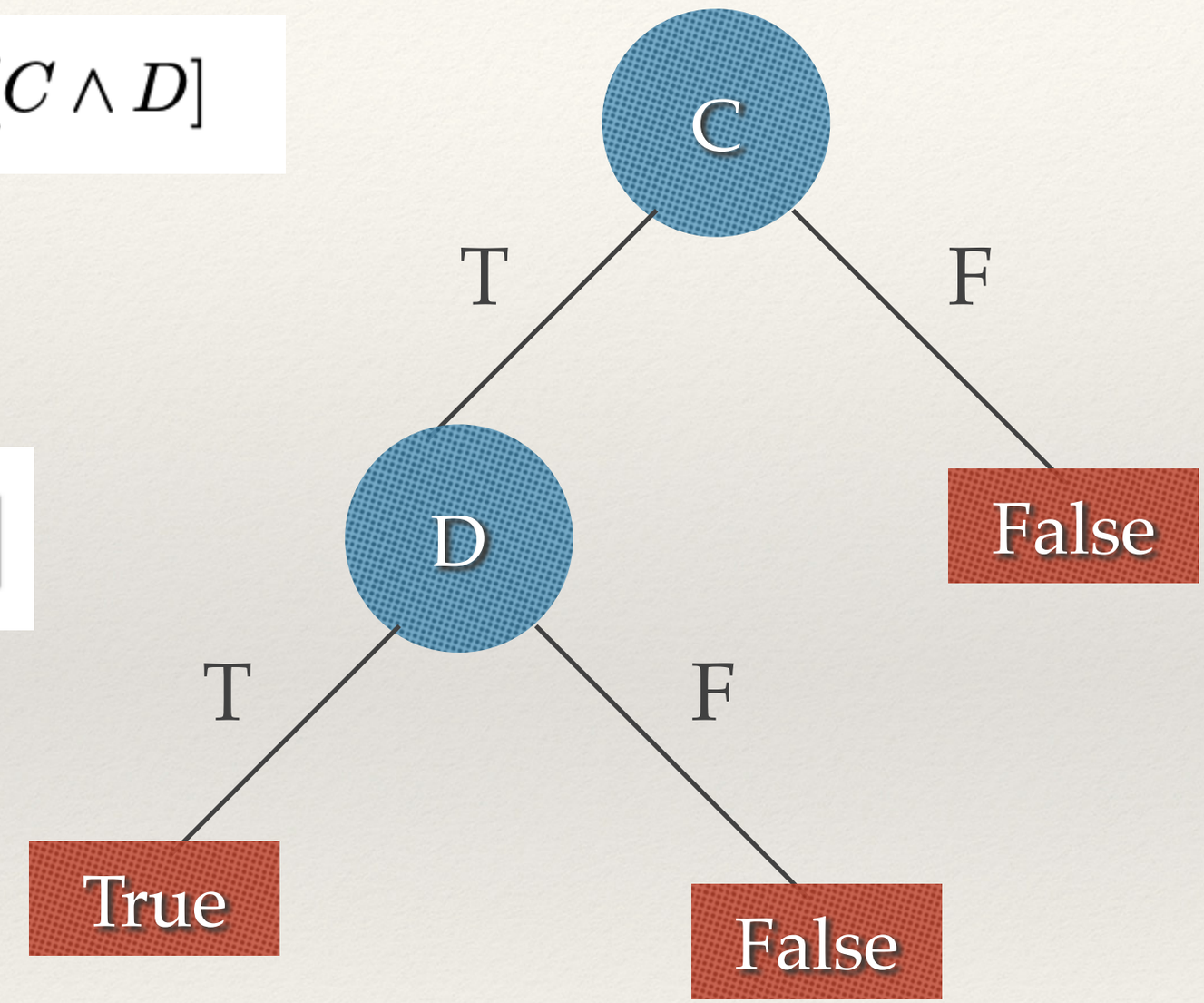
$[A \wedge B]$



Q1d Analyze

(d) $[A \wedge B] \vee [C \wedge D]$

$[C \wedge D]$

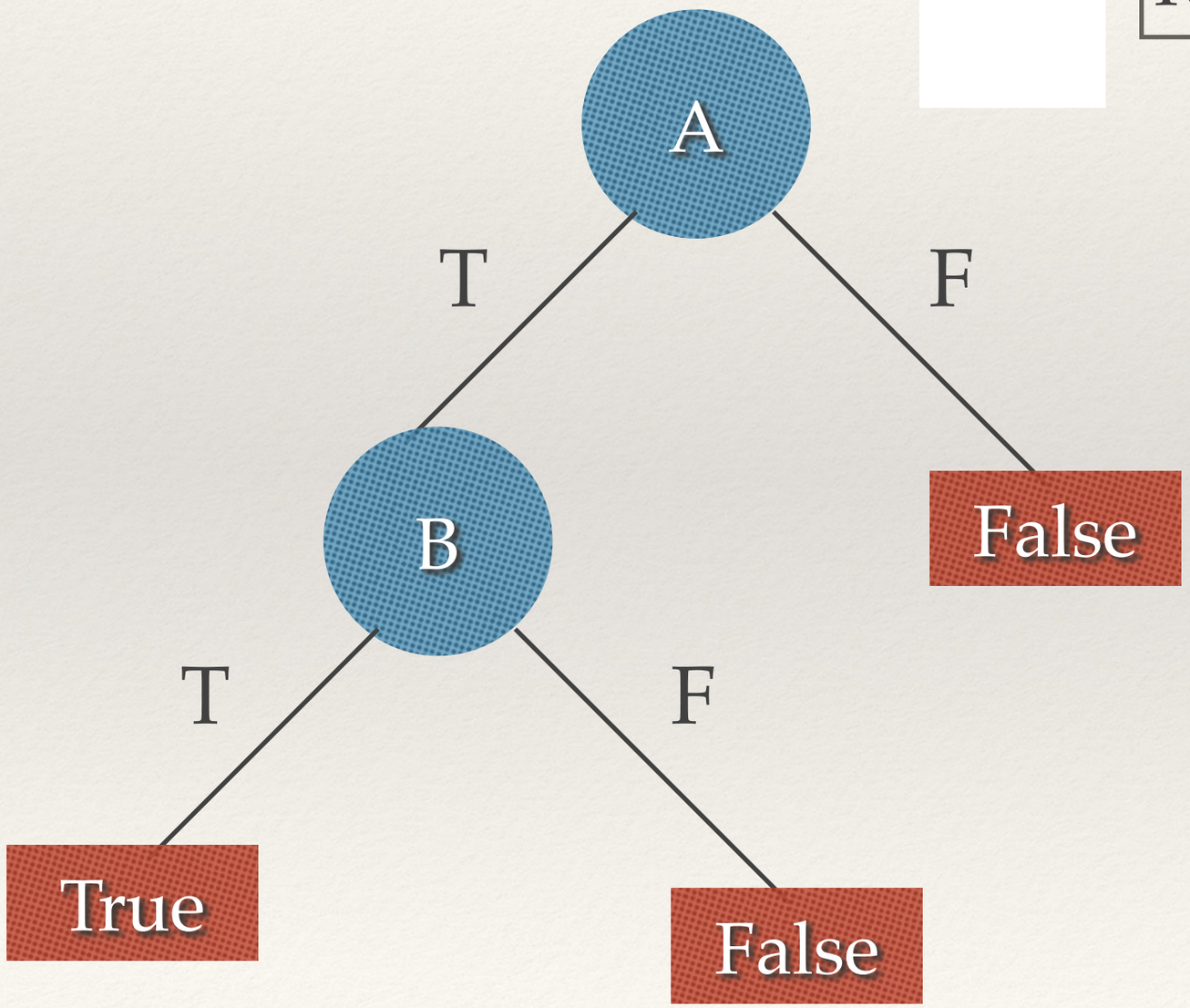


Q1d Analyze

(d) $[A \wedge B] \vee [C \wedge D]$

$\vee$

Replace the False Leaf Nodes.

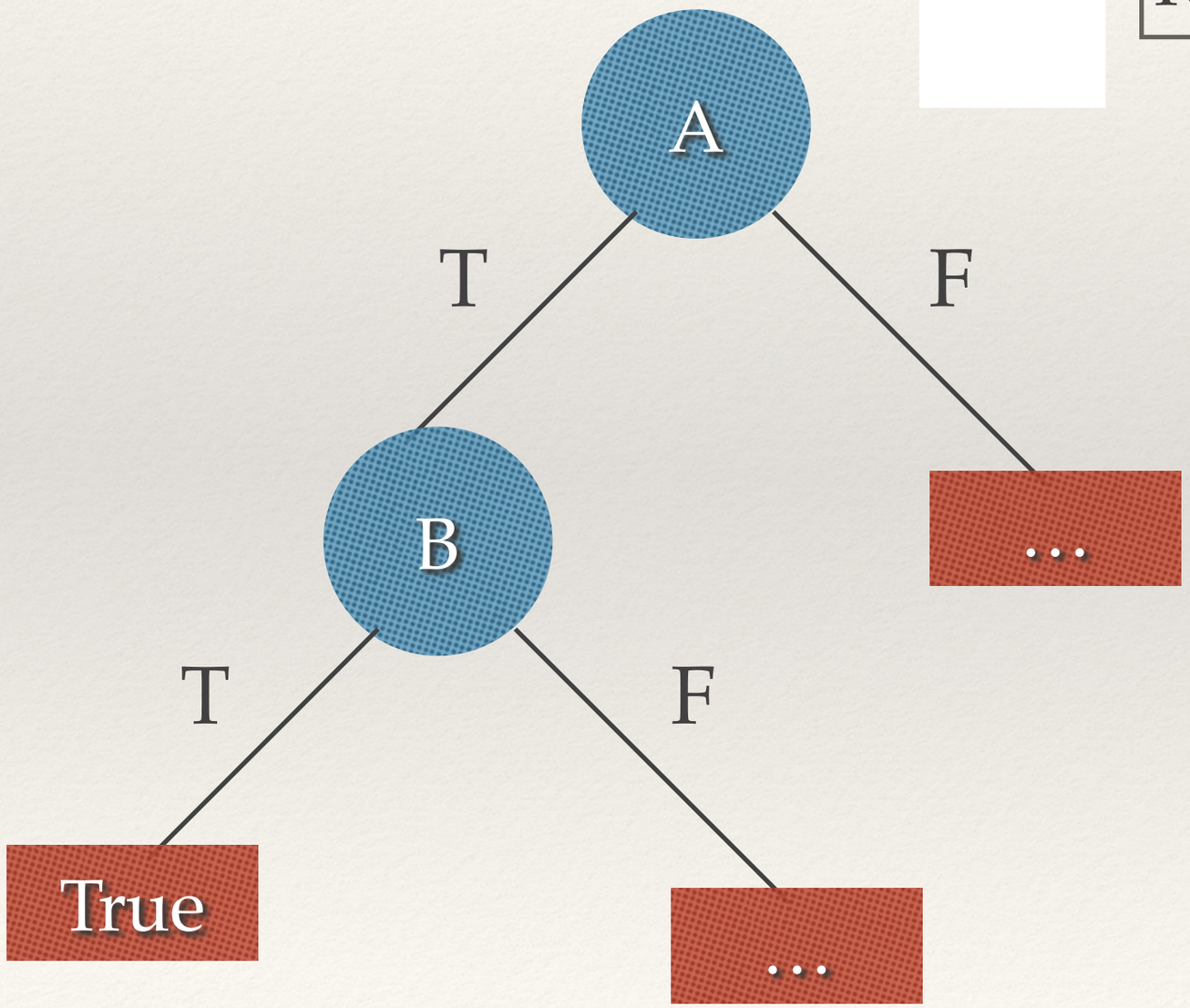


Q1d Analyze

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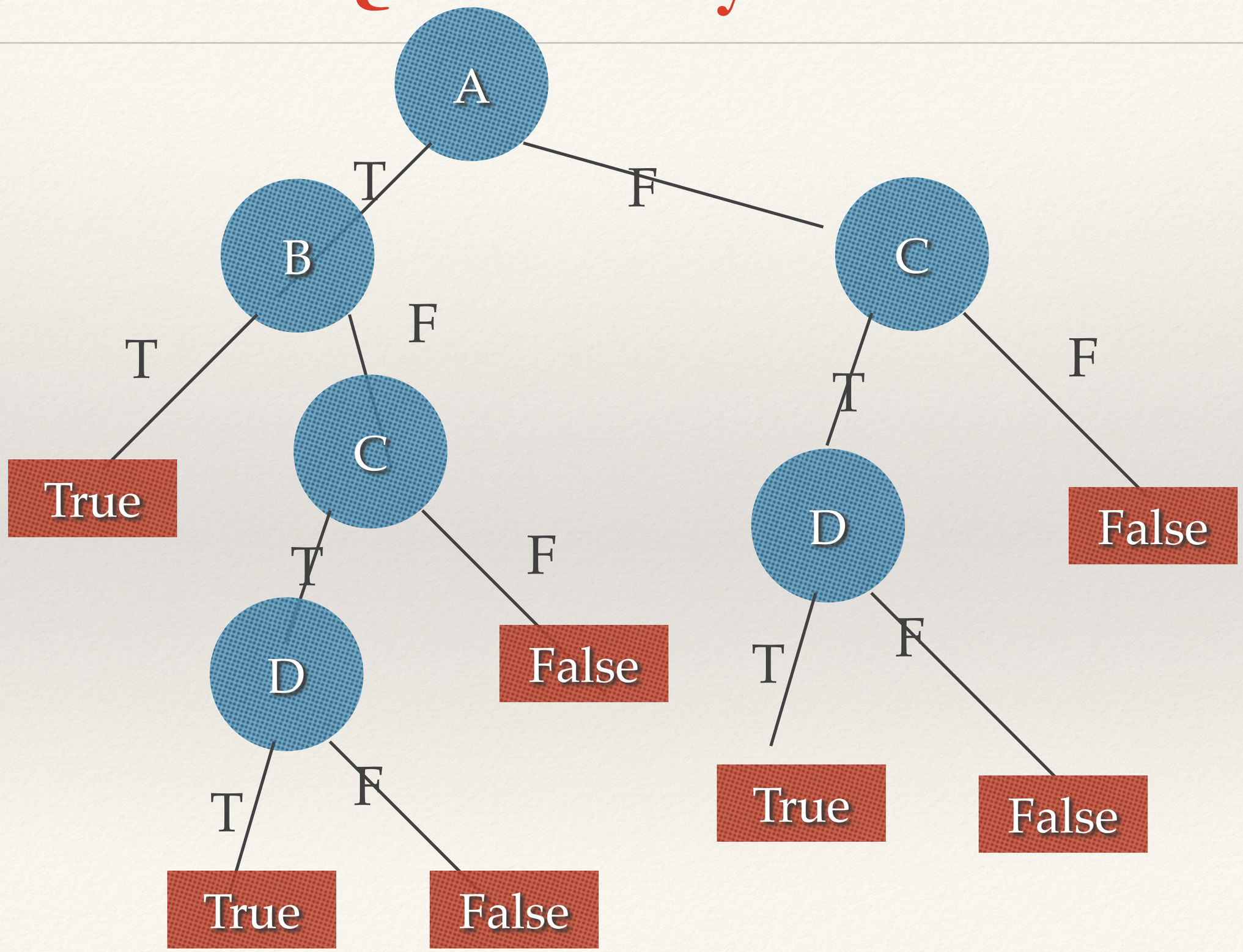
$\vee$

Replace the False Leaf Nodes.



(d) $[A \wedge B] \vee [C \wedge D]$

Q1d Analyze



Q1d Analyze

(d) $[A \wedge B] \vee [C \wedge D]$

```
A = t:
|     B = t:   True
|     B = f:
|     |     C = t:
|     |     |     D = t:   True
|     |     |     D = f:   False
|     |     C = f:   False
A = f:
|     C = t:
|     |     D = t:   True
|     |     D = f:   False
|     C = f:   False
```

Question 2

Question 2. Decision Tree Learning

(a) Assume we learn a decision tree to predict class Y given attributes A, B and C from the following training set, with no pruning.

A	B	C	Y
0	0	0	0
0	0	1	0
0	0	1	0
0	1	0	0
0	1	1	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	0	1
1	1	1	0
1	1	1	1

What would be the training set error for this dataset ? Express your answer as the number of examples out of twelve that would be misclassified.

(b) One nice feature of decision tree learners is that they can learn trees to do *multi-class* classification, i.e., where the problem is to learn to classify each instance into exactly one of $k > 2$ classes.

Q2a Analyze

Question 2. Decision Tree Learning

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0	0	1	0
0	1	0	0
0	1	1	0
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Deterministic

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Q2a Analyze

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0	1	1	1
1	0	0	0
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1	1	0	1
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Nondeterministic

What would be the training set error for this dataset ? Express your answer as the number of examples out of twelve that would be misclassified.

Q2a Analyze

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1	0	0	0
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Deterministic

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Q2a Analyze

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1	1	1	0
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Nondeterministic

What would be the training set error for this dataset ? Express your answer as the number of examples out of twelve that would be misclassified.

$$2/12 = 16.67\%$$

Q2b

(b) One nice feature of decision tree learners is that they can learn trees to do *multi-class* classification, i.e., where the problem is to learn to classify each instance into exactly one of $k > 2$ classes.

Suppose a decision tree is to be learned on an arbitrary set of data where each instance has a discrete class value in one of $k > 2$ classes. What is the maximum training set error, expressed as a fraction, that any dataset could have ?

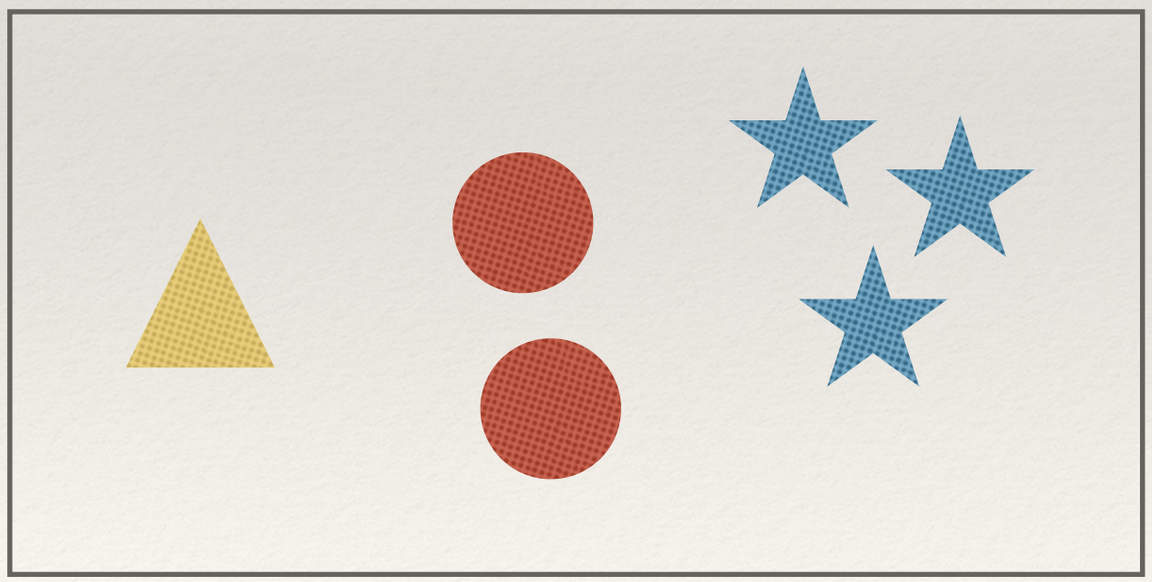
Q2b Analyze

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Consider Each Leaf Node:

k = 3 classes



Prediction



3 True / 6 instances

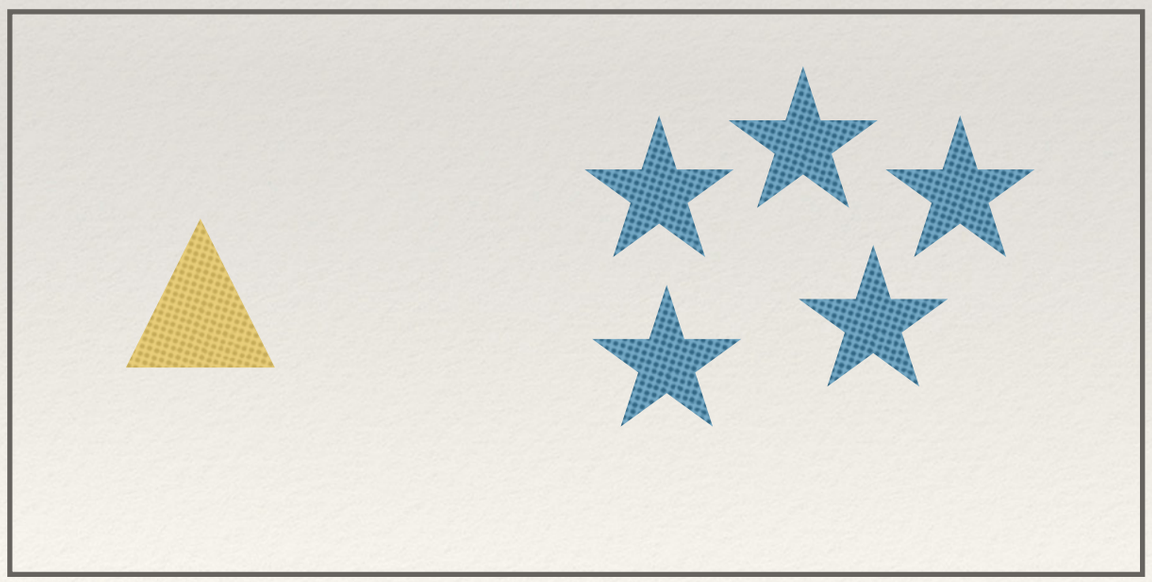
Q2b Analyze

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Consider Each Leaf Node:

k = 3 classes



Prediction



5 True / 6 instances

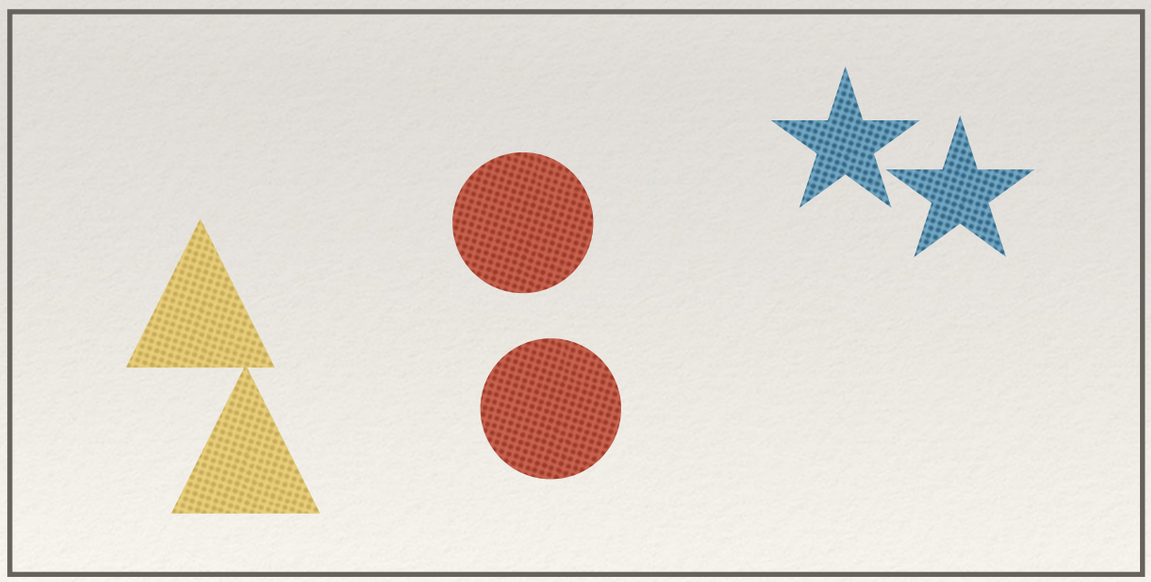
Q2b Analyze

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Consider Each Leaf Node:

k = 3 classes



Prediction



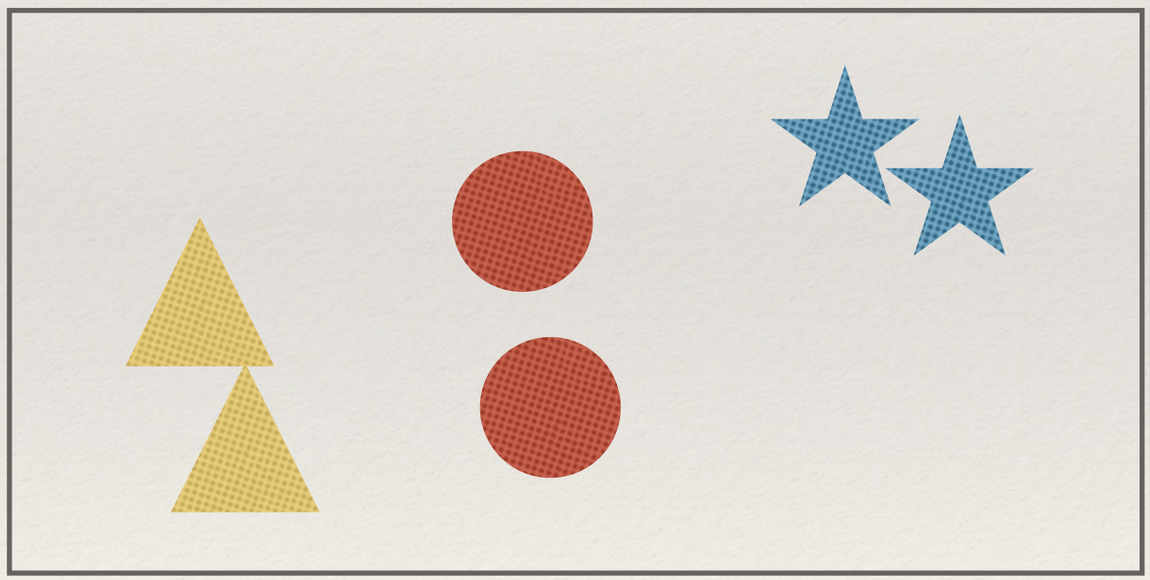
2 True / 6 instances

Q2b Analyze

(b) One nice feature of decision tree learners is that they can learn trees to do *multi-class* classification, i.e., where the problem is to learn to classify each instance into exactly one of $k > 2$ classes.

Suppose a decision tree is to be learned on an arbitrary set of data where each instance has a discrete class value in one of $k > 2$ classes. What is the maximum training set error, expressed as a fraction, that any dataset could have ?

Consider Each Leaf Node: k = 3 classes

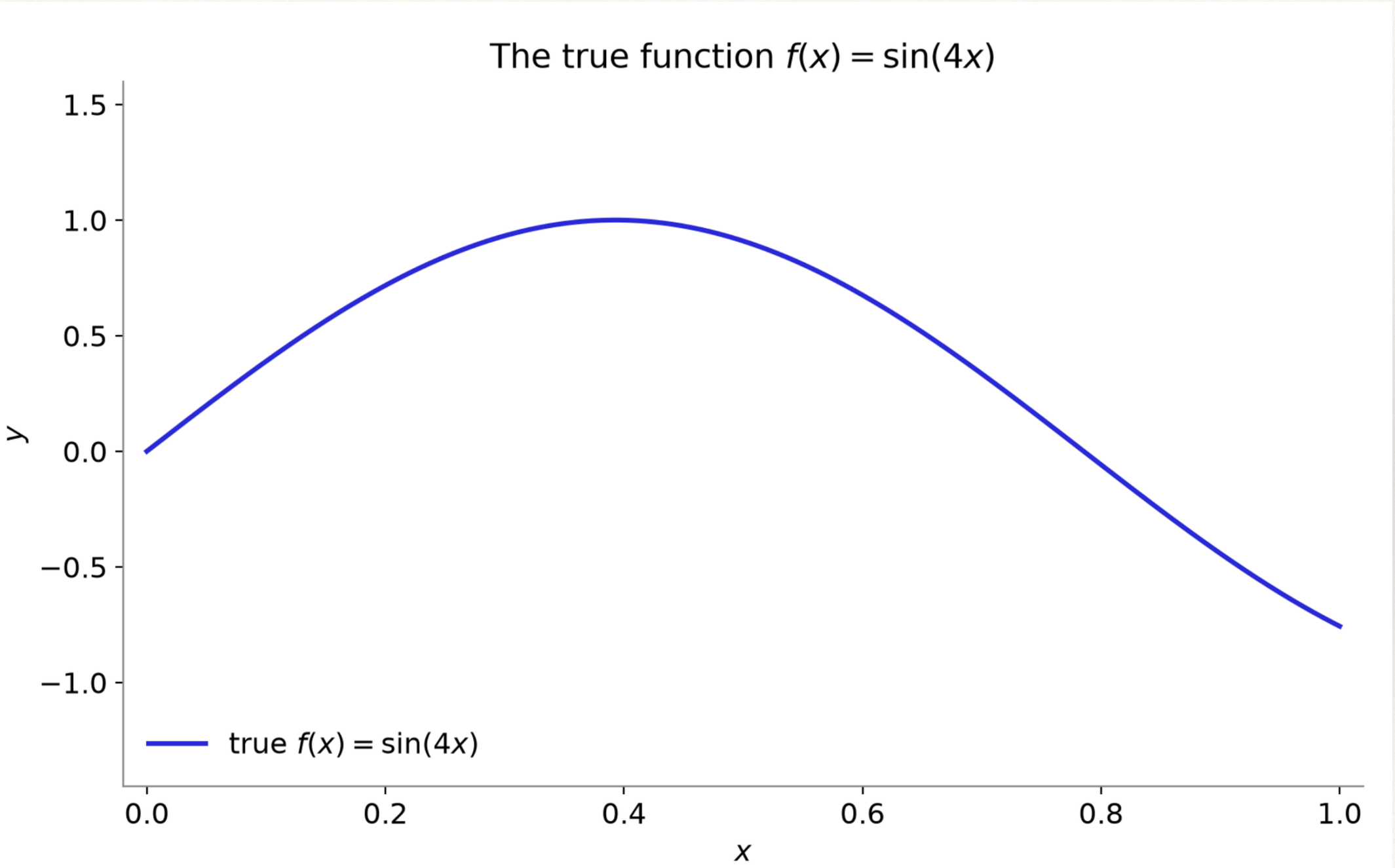


Maximum Training Error: (k-1)/k

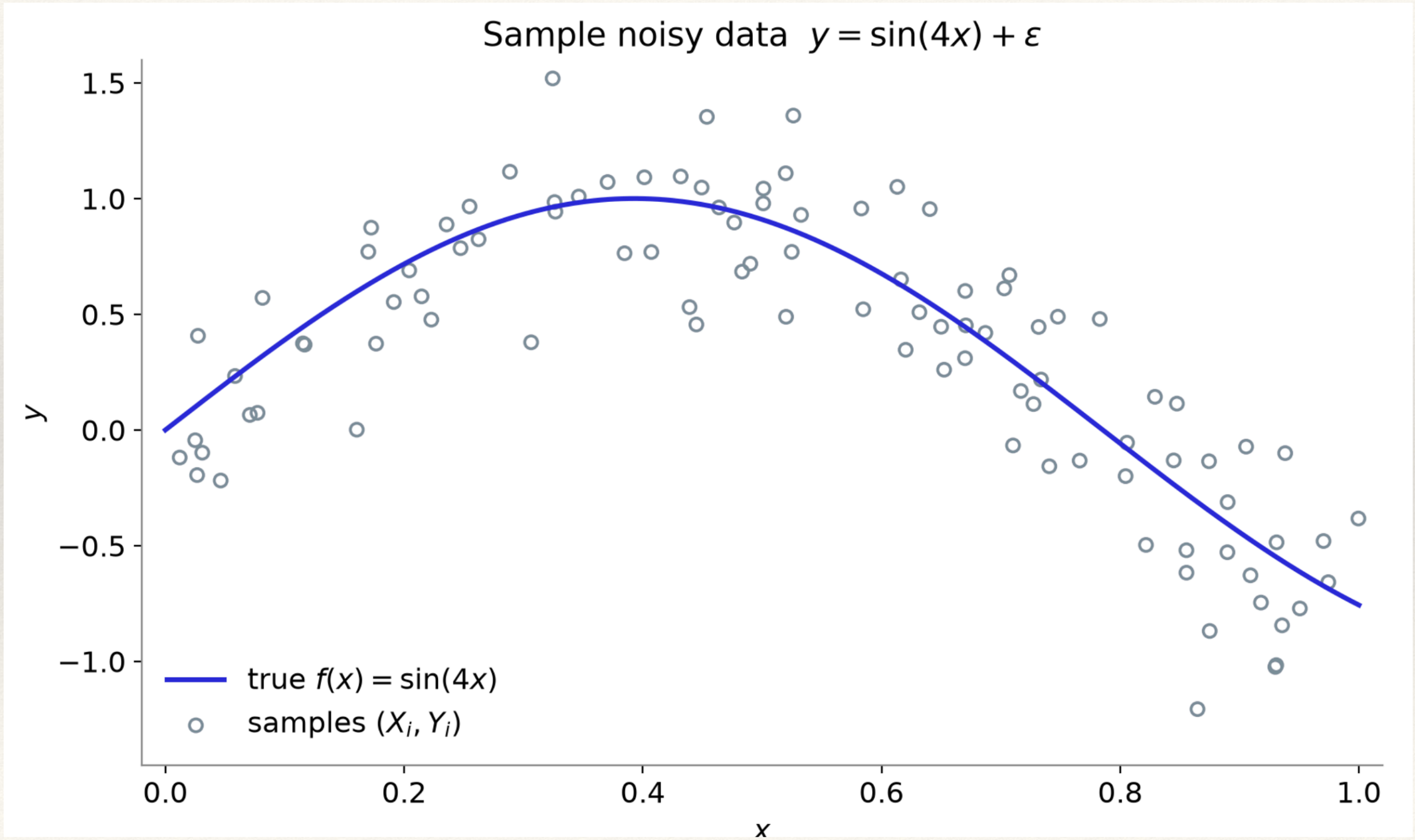
The node contains an equal mix of all k classes.

Question 3

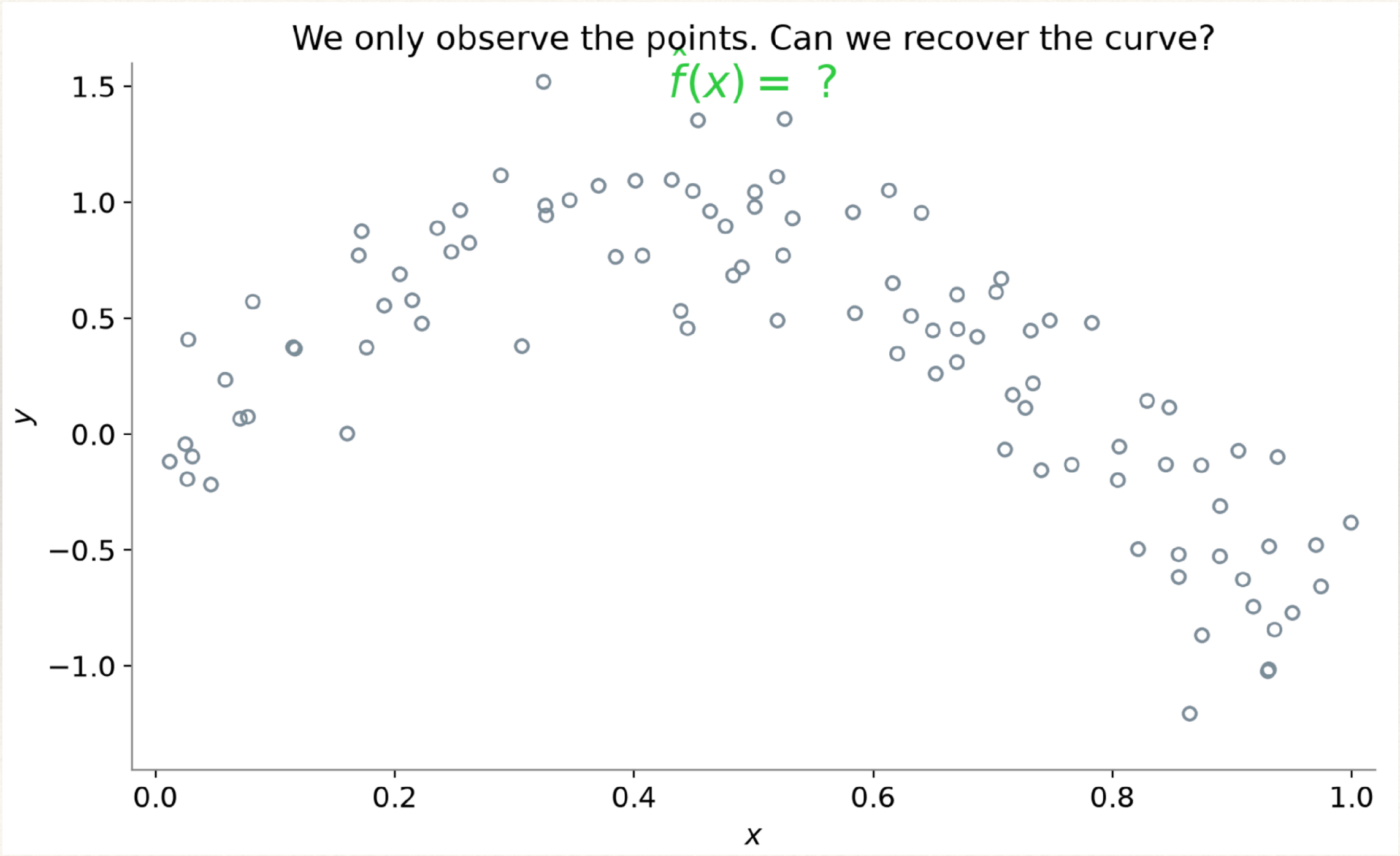
linear smoothing



Question 3

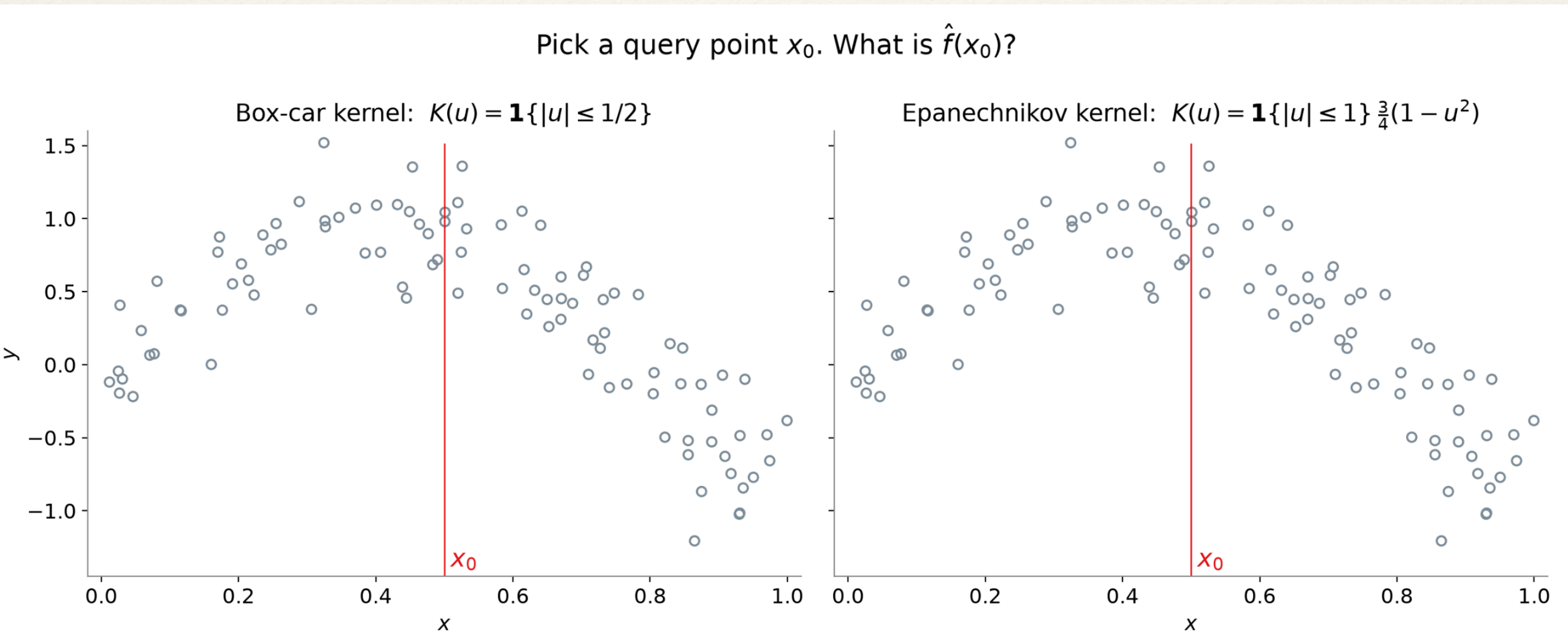

Sample noisy data $y = \sin(4x) + \varepsilon$
1.5
1.0
0.5
0.0
−0.5
−1.0
y
0.0
0.2
0.4
0.6
0.8
1.0
x
true $f(x) = \sin(4x)$
samples (X_i, Y_i)

Question 3


We only observe the points. Can we recover the curve?
$\hat{f}(x) = ?$
1.5
1.0
0.5
0.0
−0.5
−1.0
y
0.0
0.2
0.4
0.6
0.8
1.0
x

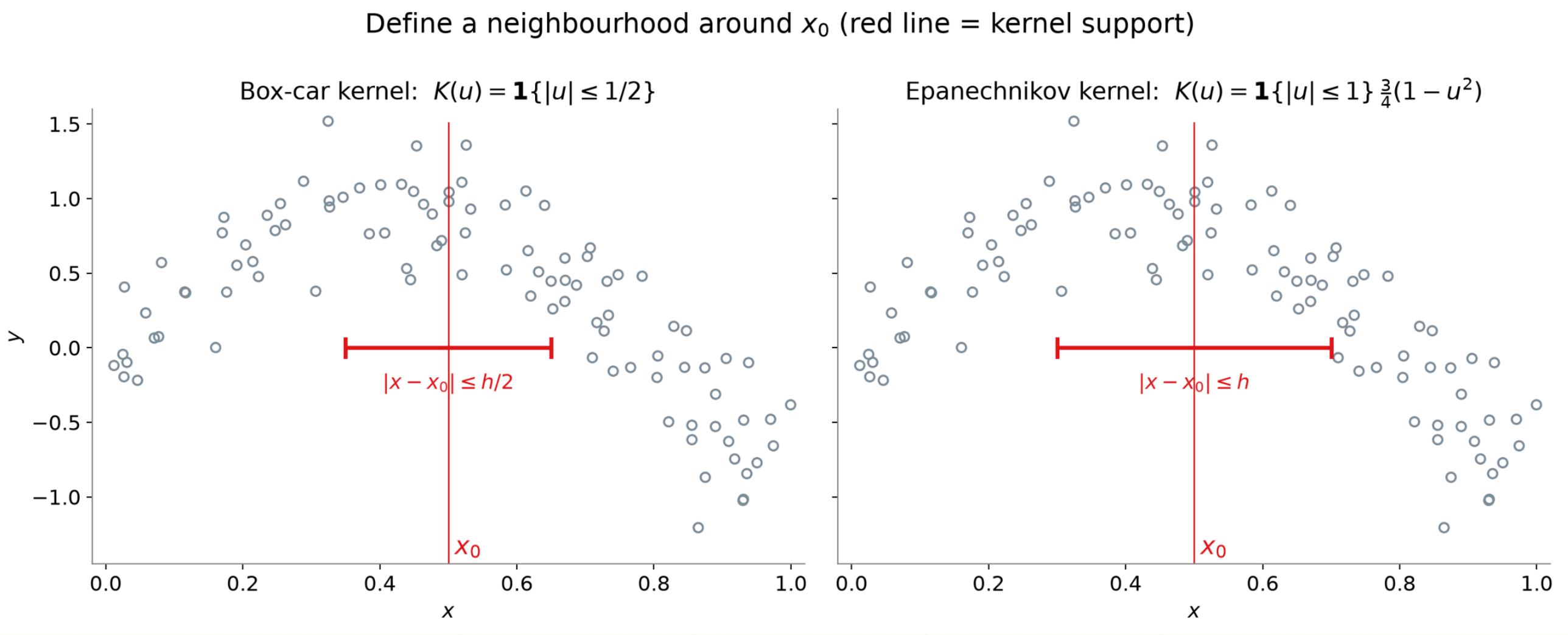
Q3 Analyze

Consider each point:



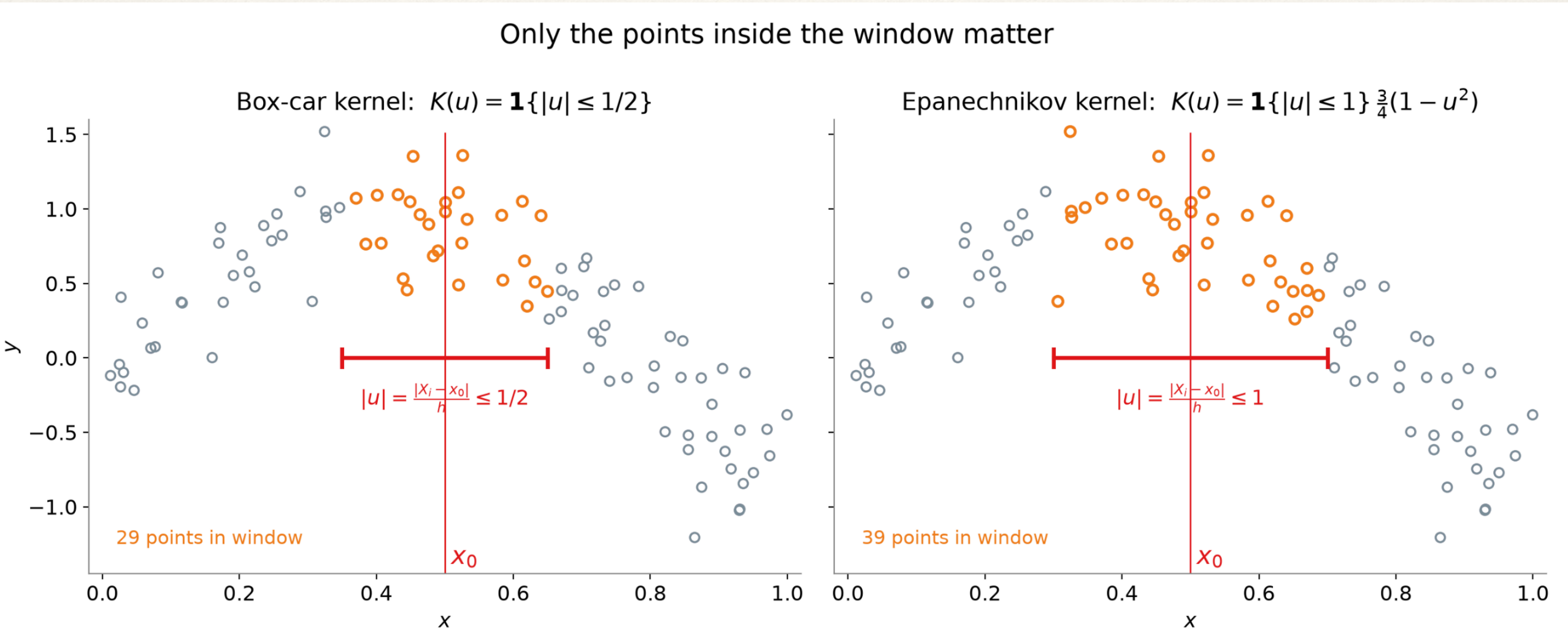
Q3 Analyze

Consider each point:



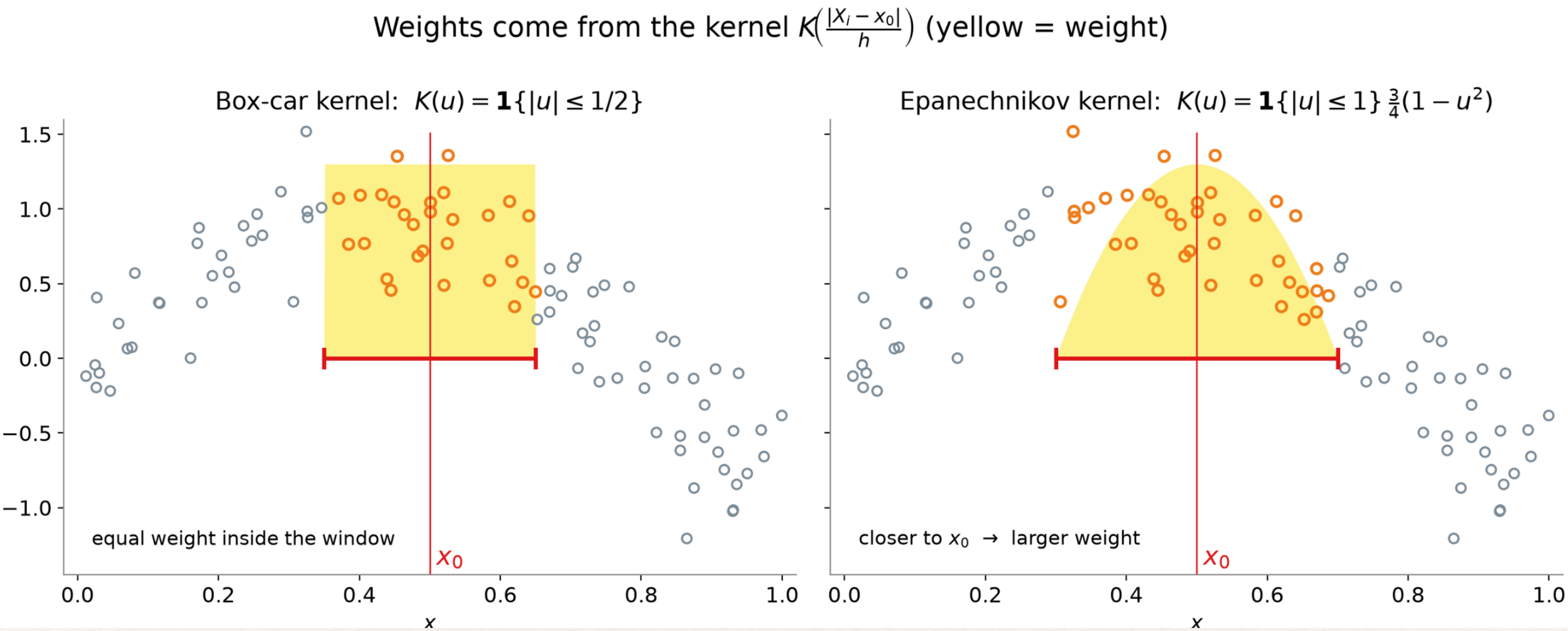
Q3 Analyze

Consider each point:



Q3 Analyze

Consider each point:

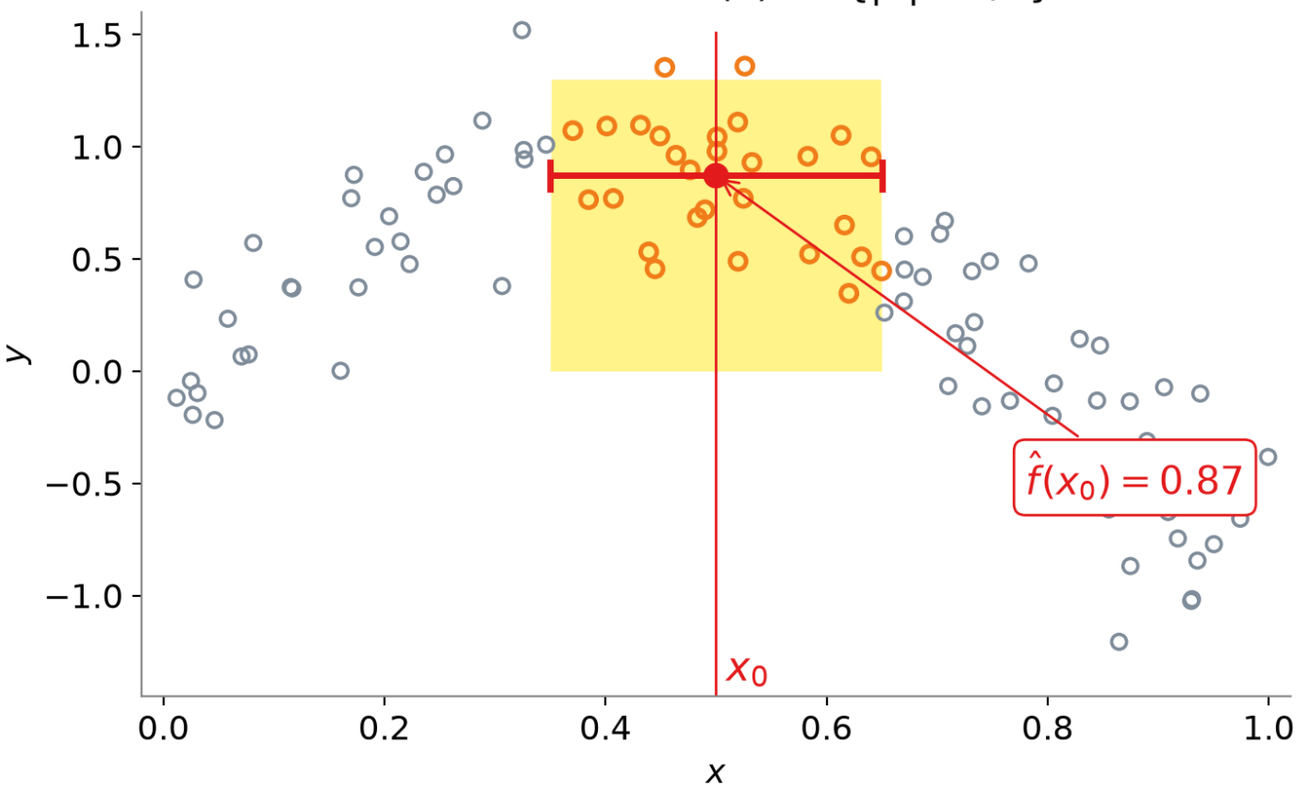


Q3 Analyze

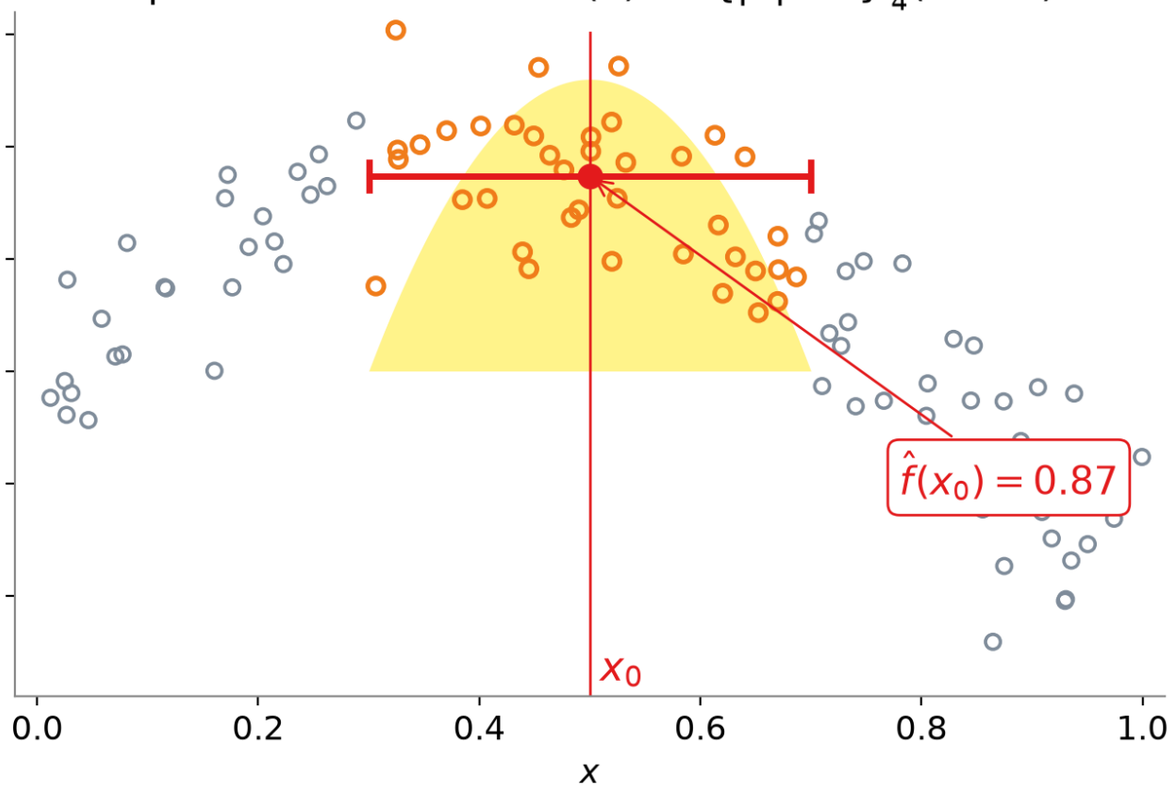
Consider each point:

Weighted average $\hat{f}(x_0) = \sum_i w_i(x_0)\, Y_i$

Box-car kernel: $K(u) = \mathbf{1}\{|u| \le 1/2\}$

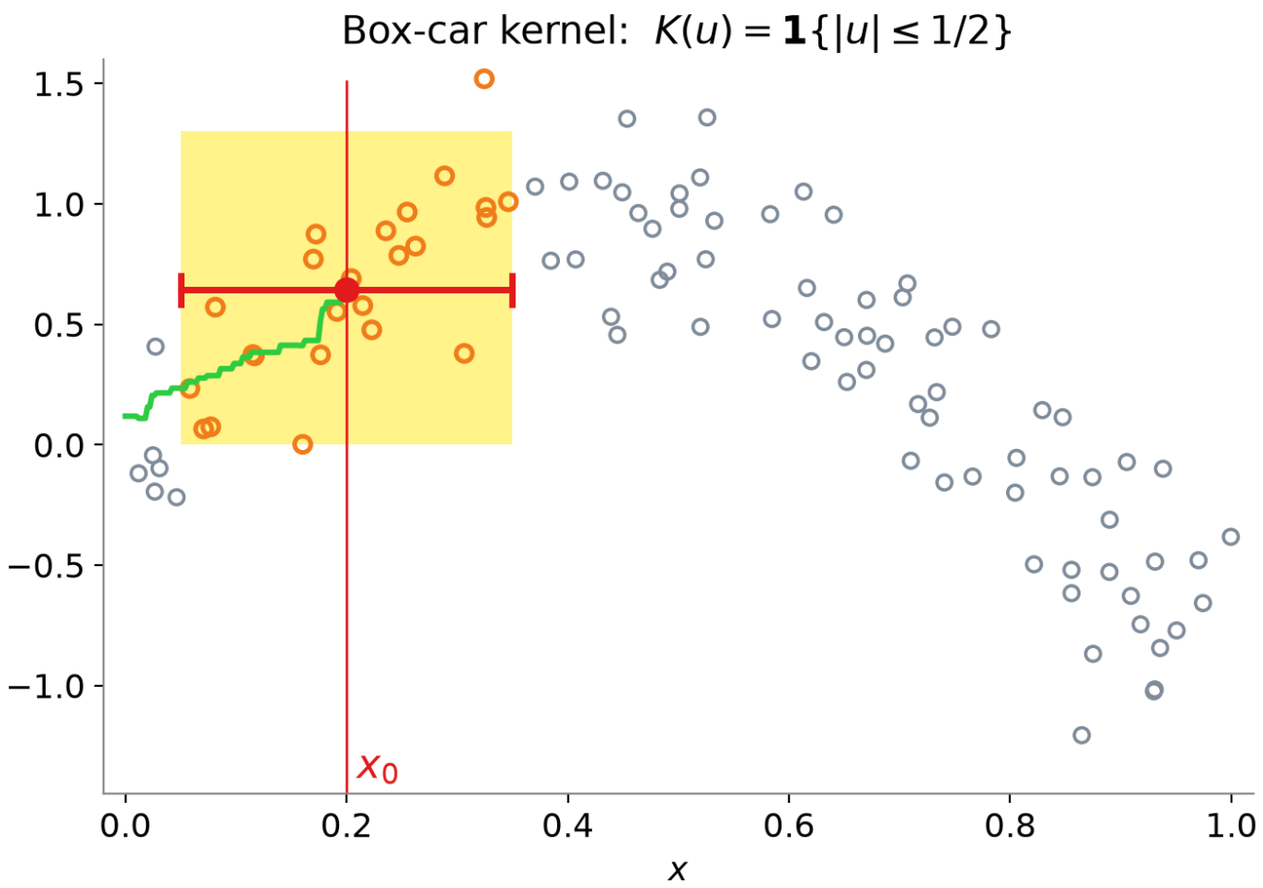


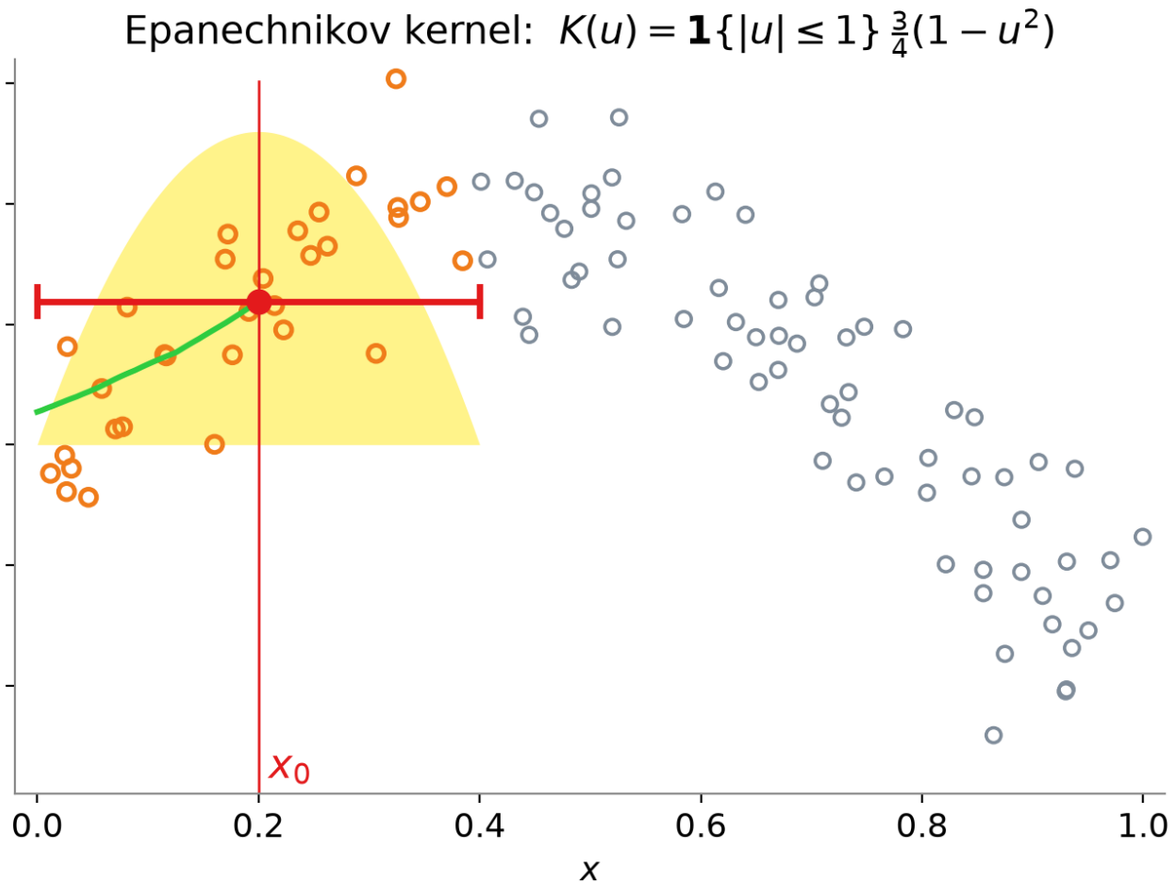
Epanechnikov kernel: $K(u) = \mathbf{1}\{|u| \le 1\}\, \frac{3}{4}(1-u^2)$



Q3 Analyze

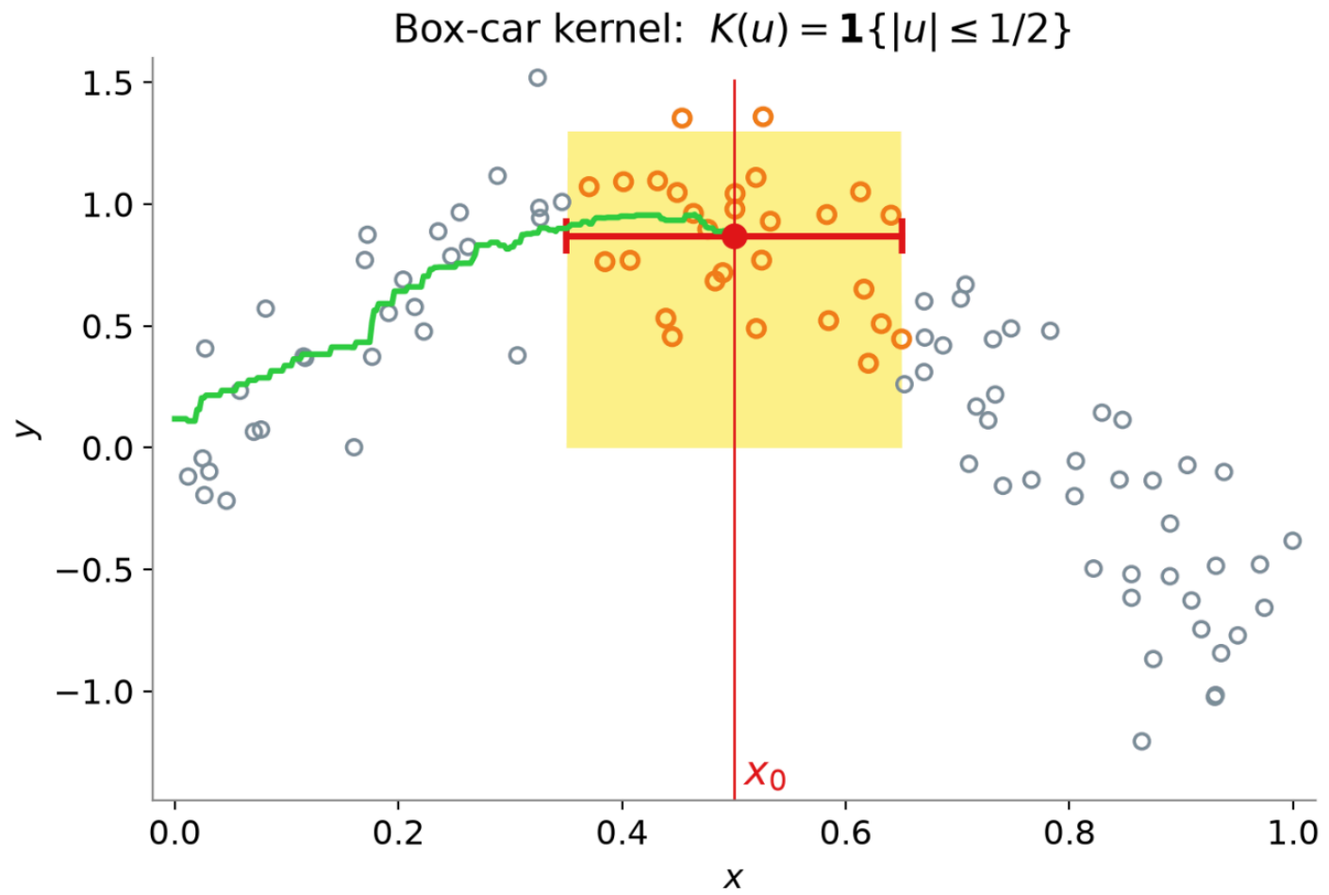
Repeat this for each point:

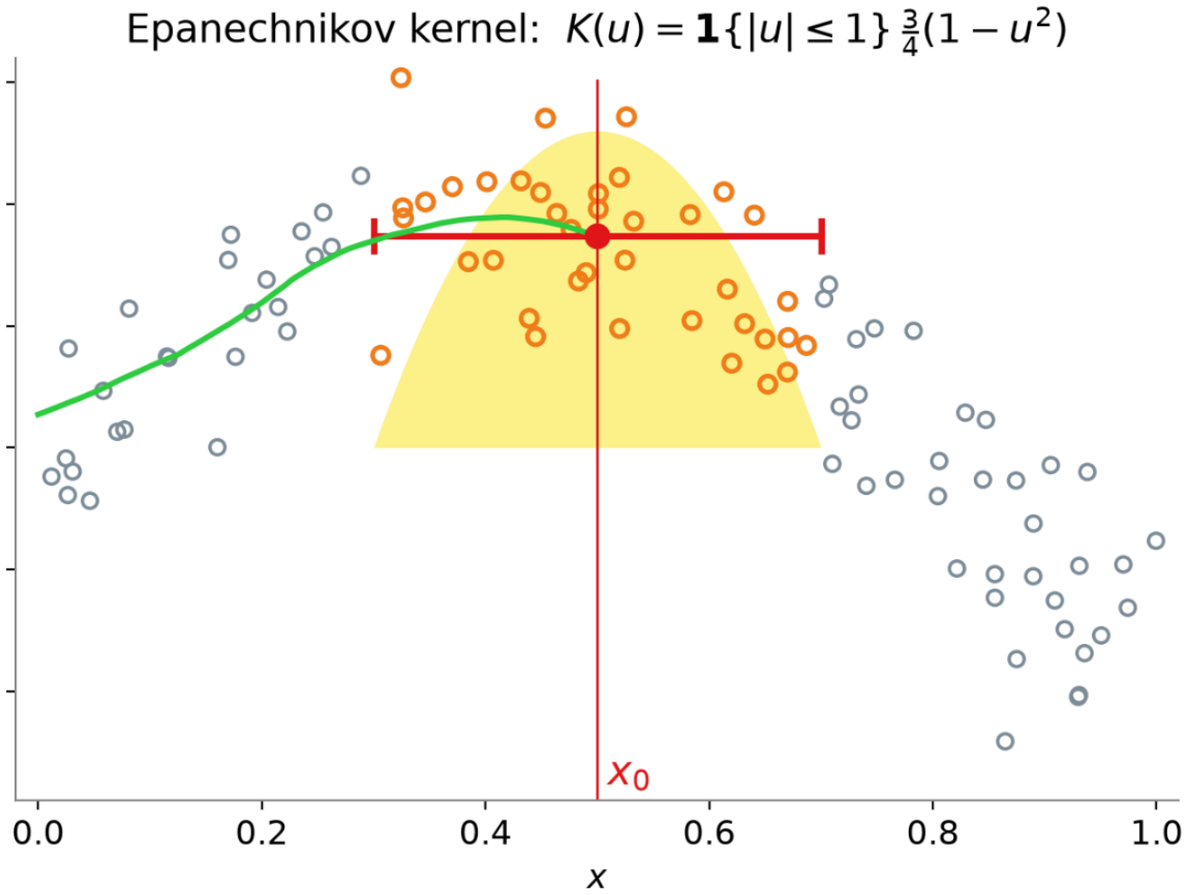




Q3 Analyze

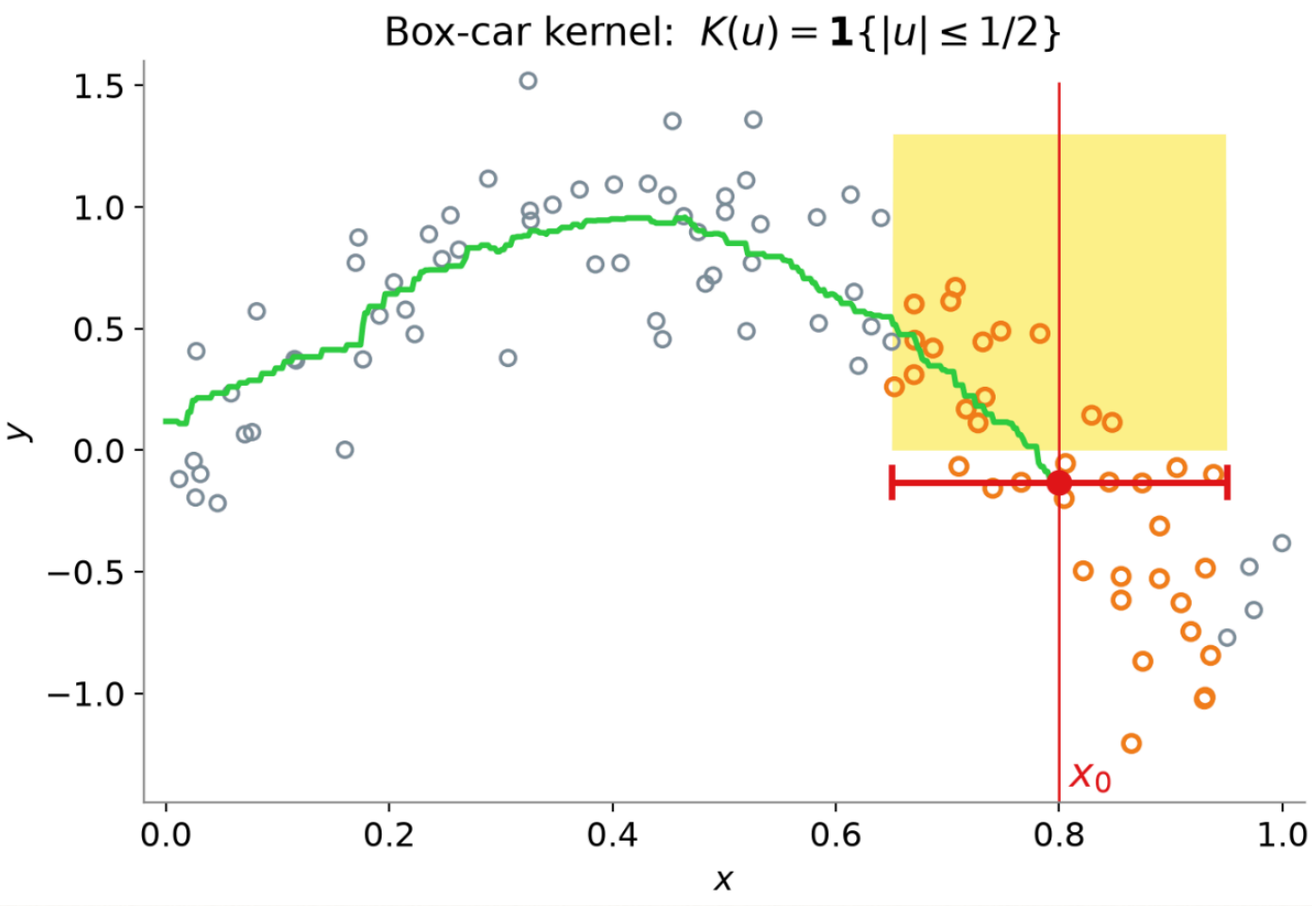
Repeat this for each point:

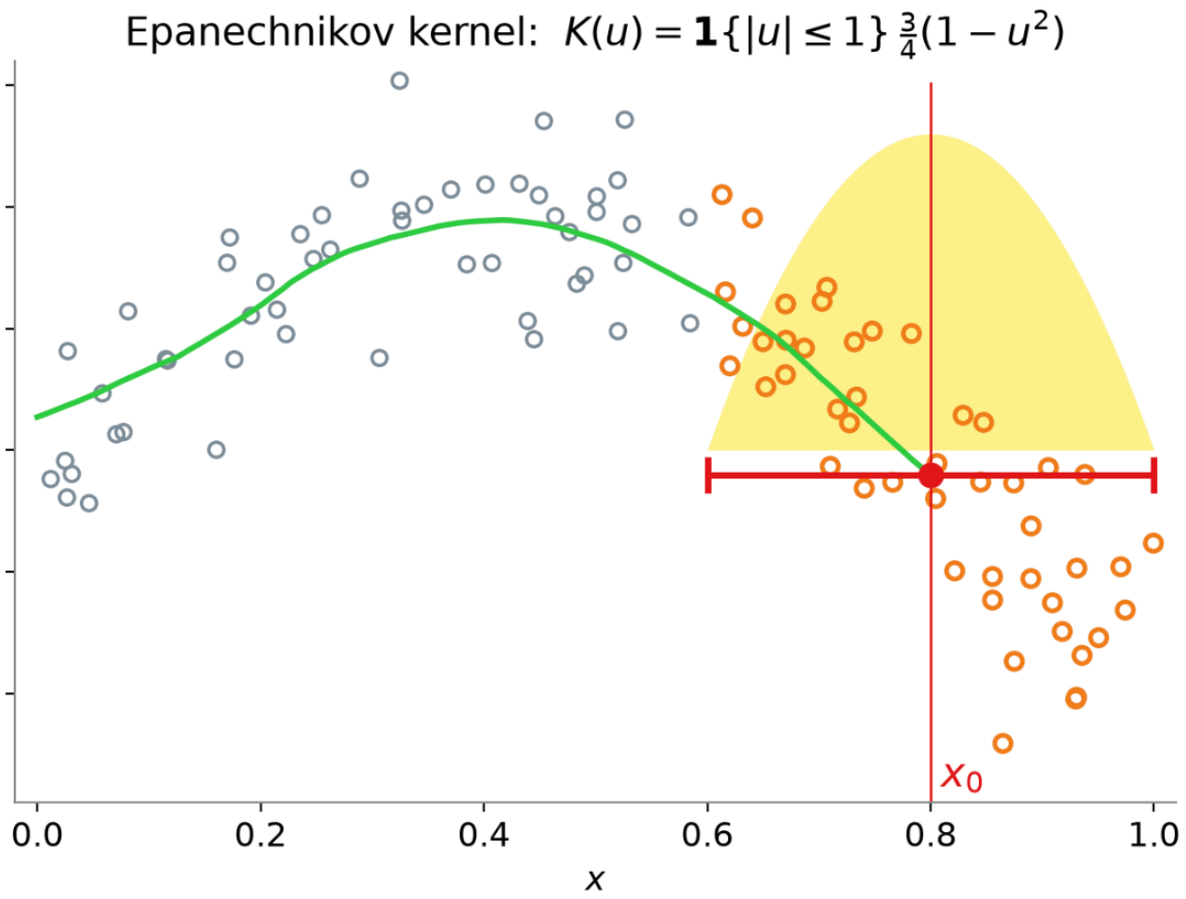




Q3 Analyze

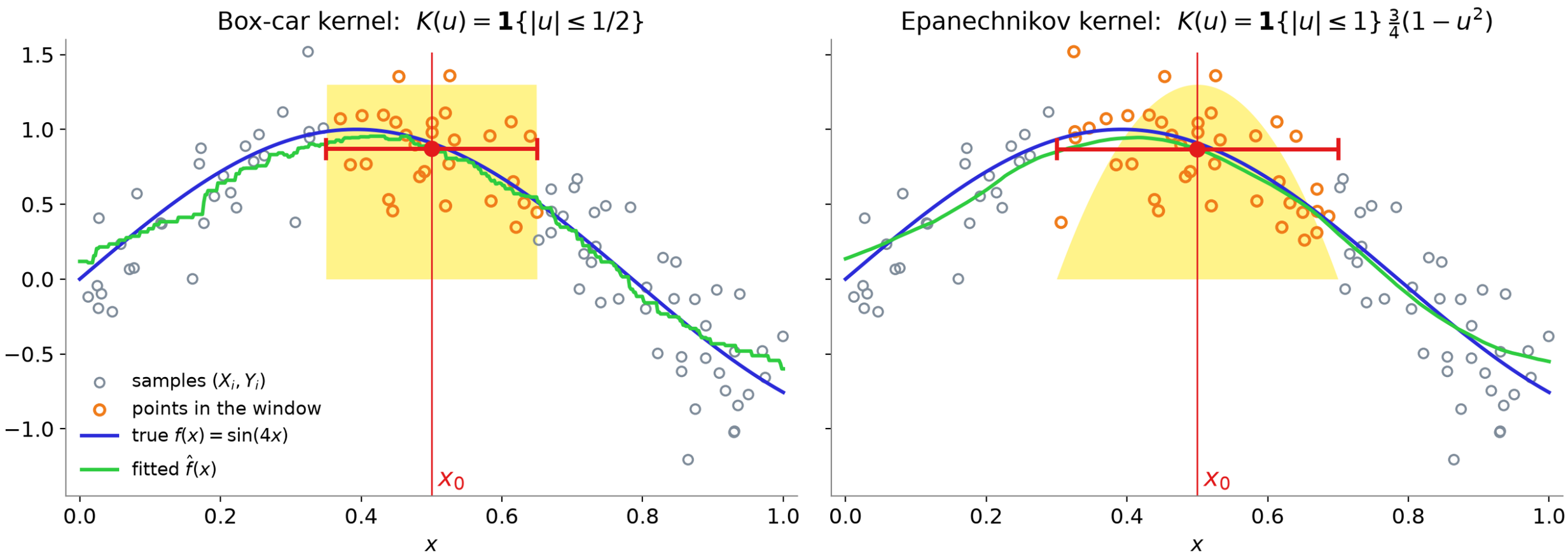
Repeat this for each point:





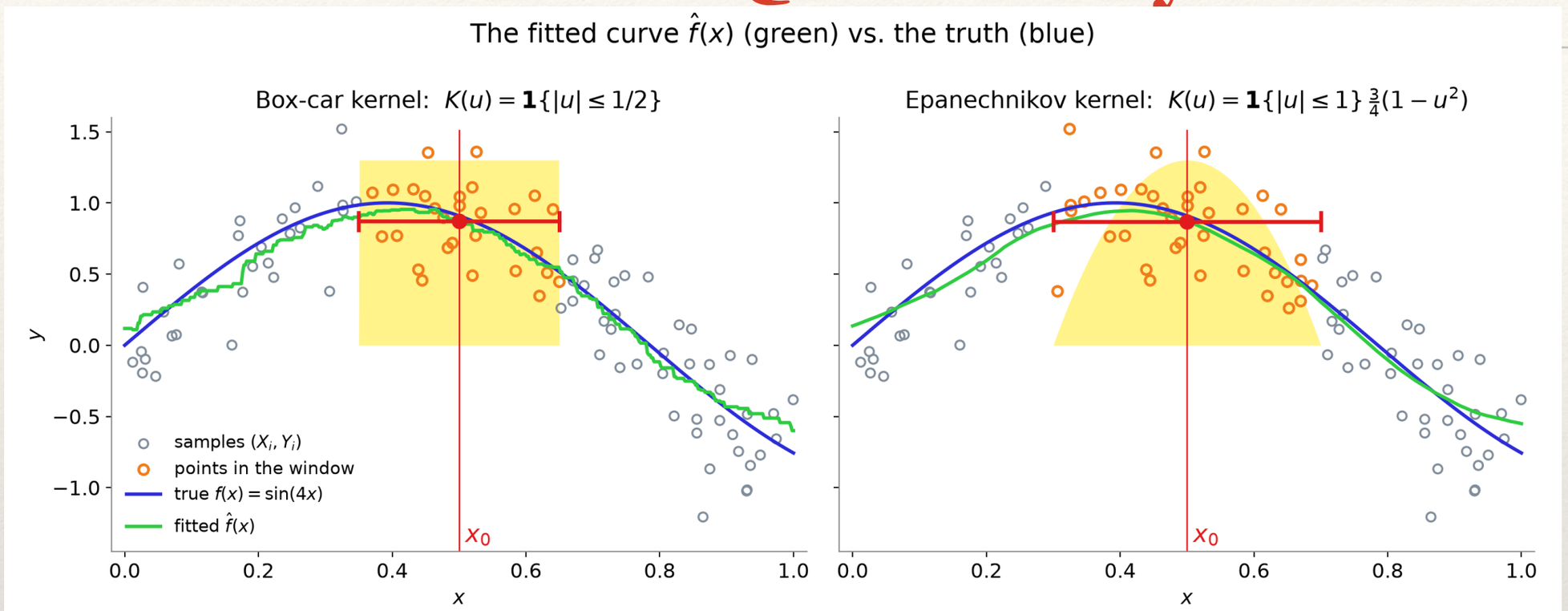
Q3 Analyze

The fitted curve $\hat{f}(x)$ (green) vs. the truth (blue)



(i) the blue curve
(ii) the black scatter
(iii) the red scatter
(iv) the yellow region
(v) the horizontal red line
(vi) the red point on the horizontal red line
(vii) the green curve

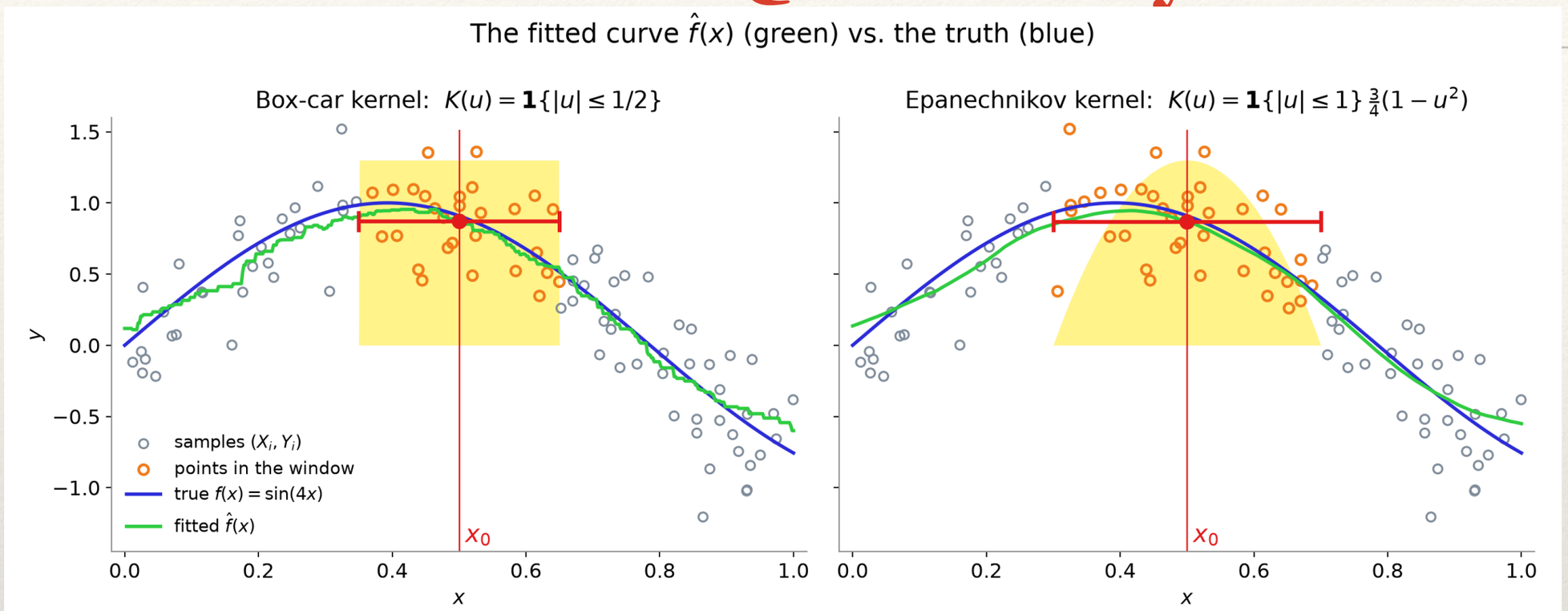
Q3 Analyze



(i) the blue curve
(ii) the black scatter
(iii) the red scatter
(iv) the yellow region
(v) the horizontal red line
(vi) the red point on the horizontal red line
(vii) the green curve

(i) The true function
(ii) The noised sample/observation
(iii) the points that meet the conditons of the kernel
(iv) The weight/kernel function.
(v) The neighbourhood/window

Q3 Analyze



(i) the blue curve
(ii) the black scatter
(iii) the red scatter
(iv) the yellow region
(v) the horizontal red line
(vi) the red point on the horizontal red line
(vii) the green curve

(vi) The Prediction for x0

(vii) The fitted function

Thank you!